TRIGONOMETRY

त्रिकोणमिति

SHEET-03

CLASS NOTES

TRIGONOMETRIC IDENTITIES

1. **Select the INCORRECT formula from the following options.**

निम्नलिखित विकल्पों में से गलत सूत्र का चयन कीजिए।

SSC CGL 17/07/2023 (Shift-01)

(a) $\sec^2\theta - \tan^2\theta = 1$

(b) $\sin^2\theta + \cos^2\theta = 1$

(c) $\text{cosec}^2\theta - \cot^2\theta = 1$

(d) $\sec^2\theta + \cos^2\theta = 1$

$\theta = 45^\circ$

(a) $2 - 1 = 1$

(b) $\frac{1}{2} + \frac{1}{2} = 1$

(c) $2 - 1 = 1$

(d) $2 + \frac{1}{2} = \frac{5}{2}$

1. Select the INCORRECT formula from the following options.

निम्नलिखित विकल्पों में से गलत सूत्र का चयन कीजिए।

SSC CGL 17/07/2023 (Shift-01)

(a) $\sec^2\theta - \tan^2\theta = 1$

(b) $\sin^2\theta + \cos^2\theta = 1$

(c) $\text{cosec}^2\theta - \cot^2\theta = 1$

(d) $\sec^2\theta + \cos^2\theta = 1$

ⓐ $\frac{25}{16} - \frac{9}{16} = \frac{16}{16} = 1$

ⓑ $\frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1$

ⓒ $\frac{25}{9} - \frac{16}{9} = \frac{9}{9} = 1$

ⓓ $\frac{25}{16} + \frac{16}{25} \neq 1$

1. Select the INCORRECT formula from the following options.

निम्नलिखित विकल्पों में से गलत सूत्र का चयन कीजिए।

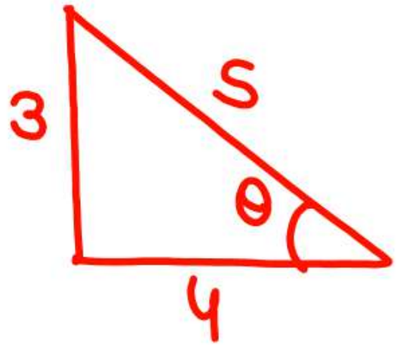
SSC CGL 17/07/2023 (Shift-01)

(a) $\sec^2\theta - \tan^2\theta = 1$

(b) $\sin^2\theta + \cos^2\theta = 1$

(c) $\text{cosec}^2\theta - \cot^2\theta = 1$

(d) $\sec^2\theta + \cos^2\theta = 1$



- $\tan\theta + \cot\theta$

Proof:-

$$\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}$$

$$= \frac{\sin^2\theta + \cos^2\theta}{\sin\theta \cdot \cos\theta}$$

$$= \frac{1}{\sin\theta \cdot \cos\theta}$$

$$= \operatorname{cosec}\theta \cdot \sec\theta$$

2. Which of the following is equal to $\frac{1}{\tan\theta} + \tan\theta$?

$\cot\theta + \tan\theta$

निम्न में से कौन $\frac{1}{\tan\theta} + \tan\theta$ के बराबर है?

SSC CGL MAINS (08/08/2022

(a) $\frac{\cos ec\,\theta}{\sec\theta}$

(b) $\tan^2\theta$

(c) $\sec\theta \cdot \operatorname{cosec}\theta$ ✓

(d) 1

$\theta = 45^\circ$

1+1 = 2

2. Which of the following is equal to $\frac{1}{\tan\theta} + \tan\theta$?

निम्न में से कौन $\frac{1}{\tan\theta} + \tan\theta$ के बराबर है?

SSC CGL MAINS (08/08/2022

(a) $\frac{\text{cosec}\,\theta}{\sec\theta}$ $\frac{\sqrt{2}}{\sqrt{2}} = 1$

(b) $\tan^2\theta$ 1

(c) $\sec\theta \cdot \text{cosec}\,\theta$

(d) 1

$\sqrt{2} \times \sqrt{2} = 2$

$\frac{4}{3}+\frac{3}{4}=\frac{25}{12}$

2. Which of the following is equal to $\frac{1}{\tan\theta}+\tan\theta$?

निम्न में से कौन $\frac{1}{\tan\theta}+\tan\theta$ के बराबर है?

SSC CGL MAINS (08/08/2022

$\frac{20}{15}=\left(\frac{5}{3}\right)\left(\frac{5}{4}\right)$

(a) $\frac{\text{cosec}\,\theta}{\sec\theta}$

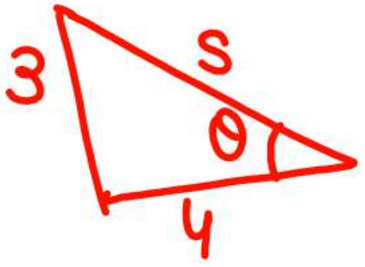
(b) $\tan^2\theta$ $\frac{9}{16}$

(c) $\sec\theta \cdot \text{cosec}\,\theta$

(d) 1

$\frac{5}{4}\times\frac{5}{3}$

$=\frac{25}{12}$



3. Which of the following is equal to $\left[\frac{\cos\theta}{\sin\theta}+\frac{\sin\theta}{\cos\theta}\right]$?

cotθ + tanθ

निम्न में से कौन $\left[\frac{\cos\theta}{\sin\theta}+\frac{\sin\theta}{\cos\theta}\right]$ के बराबर है?

SSC CGL MAINS (08/08/2022)

(a) $\cot\theta \cdot \sec\theta$
(b) $\operatorname{cosec}\theta \cdot \sec\theta$
(c) $\sec\theta \cdot \tan\theta$
(d) $\operatorname{cosec}\theta \cdot \tan\theta$

4. Which of the following is equal to secA – cosA?

निम्न में से कौन **secA – cosA** के बराबर है?

SSC CGL MAINS (08/08/2022)

(a) sinA. cotA **(b) cotA. cosA**

(c) tanA. sinA **(d) cosA sinA**

$$\frac{1}{c} - c$$

$$= \frac{1-c^2}{c} = \frac{\sin^2 A}{\cos A} = \frac{\sin A \sin A}{\cos A}$$

$$= \tan A \sin A$$

$\frac{5}{4} - \frac{4}{5} = \frac{9}{20}$

4. Which of the following is equal to secA – cosA?

निम्न में से कौन secA – cosA के बराबर है?

SSC CGL MAINS (08/08/2022)

(a) sinA. cotA **(b) cotA. cosA**

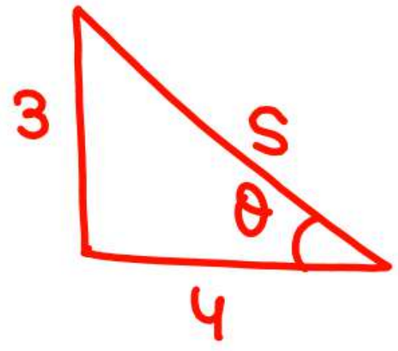
(c) tanA. sinA **(d) cosA sinA**

(a) $\frac{3}{5} \times \frac{4}{3} = \frac{4}{5}$

(b) $\frac{4}{5} \times \frac{4}{5} = \frac{16}{25}$

(c) $\frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$

(d) $\frac{4}{5} \times \frac{3}{5} = \frac{12}{25}$



$\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow \sec^2\theta - 1 = \tan^2\theta$

5. **[(sin θ tan θ + cos θ)² – 1] is equal to:**

[(sin θ tan θ + cos θ)² – 1] इनमें से किसके बराबर है?

SSC Phase X 02/08/2022 (Shift- 02)

(a) $\sec^2\theta$ **(b)** $\sec\theta$

(c) $\operatorname{cosec}\theta$ **(d)** $\tan^2\theta$

$$\left[S \times \frac{S}{C} + C\right]^2 - 1$$

$$= \left[\frac{S^2 + C^2}{C}\right]^2 - 1$$

$$= \left[\frac{1}{C}\right]^2 - 1$$

$$= \sec^2\theta - 1$$

$$= \tan^2\theta$$

$\theta = 45^\circ$

$\left(\frac{1}{\sqrt{2}} \times 1 + \frac{1}{\sqrt{2}}\right)^2 - 1$

$= (\sqrt{2})^2 - 1$

$= 2 - 1$

$= 1$

5. **[(sin θ tan θ + cos θ)² – 1] is equal to:**

[(sin θ tan θ + cos θ)² – 1] इनमें से किसके बराबर है?

SSC Phase X 02/08/2022 (Shift- 02)

(a) $\sec^2\theta$ 2

(b) $\sec\theta$ $\sqrt{2}$

(c) $\text{cosec}\,\theta$ $\sqrt{2}$

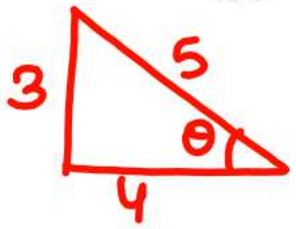
(d) $\tan^2\theta$ 1

Triplet

$\frac{5}{4} + \frac{1}{\frac{3}{4}-\frac{5}{4}}$

$= \frac{5}{4} - \frac{4}{2}$

$= \frac{5-8}{4} = -\frac{3}{4}$



6. **For $0° < \theta < 90°$, $\frac{1}{\cos\theta} + \frac{1}{\tan\theta - \sec\theta}$ is equal to:**

$0° < \theta < 90°$ के लिए, $\frac{1}{\cos\theta} + \frac{1}{\tan\theta - \sec\theta}$ बराबर है।

SSC CGL 17/08/2021 (Shift- 02)

(a) $-\sec\theta$ $-\frac{5}{4}$

~~(b) $\tan\theta$~~

~~(c) $\sec\theta$~~

(d) $-\tan\theta$ $-\frac{3}{4}$

$\theta = 45^\circ$

$$\frac{\sqrt{2}}{1} + \frac{1}{1-\sqrt{2}}$$

$$= \sqrt{2} + \frac{(1+\sqrt{2})}{1-2}$$

$$= \sqrt{2} - (1+\sqrt{2})$$

$$= \sqrt{2} - 1 - \sqrt{2}$$

$$= -1$$

6. **For $0° < \theta < 90°$, $\frac{1}{\cos\theta} + \frac{1}{\tan\theta - \sec\theta}$ is equal to:**

$0° < \theta < 90°$ के लिए, $\frac{1}{\cos\theta} + \frac{1}{\tan\theta - \sec\theta}$ बराबर है।

SSC CGL 17/08/2021 (Shift- 02)

(a) $-\sec\theta - \sqrt{2}$

(b) ~~$\tan\theta$~~

(c) ~~$\sec\theta$~~

(d) $-\tan\theta - 1$

7. The value of $\left(\frac{1}{\sin\theta}+\frac{1}{\tan\theta}\right)\left(\frac{1}{\sin\theta}-\frac{1}{\tan\theta}\right)$ is:

$\left(\frac{1}{\sin\theta}+\frac{\cos\theta}{\sin\theta}\right)\left(\frac{1}{\sin\theta}-\frac{\cos\theta}{\sin\theta}\right)$ का मान क्या होगा?

SSC CGL 18/07/2023 (Shift-02)

(a) 0 (b) 2

(c) 3 (d) 1

$\frac{(1+c)}{s}\times\frac{(1-c)}{s}=\frac{(1^2-c^2)}{s^2}=\frac{s^2}{s^2}=1$

$\theta = 45^\circ$

7. The value of $\left(\frac{1}{\sin\theta}+\frac{1}{\tan\theta}\right)\left(\frac{1}{\sin\theta}-\frac{1}{\tan\theta}\right)$ is:

$\left(\frac{1}{\sin\theta}+\frac{\cos\theta}{\sin\theta}\right)\left(\frac{1}{\sin\theta}-\frac{\cos\theta}{\sin\theta}\right)$ का मान क्या होगा?

SSC CGL 18/07/2023 (Shift-02)

(a) 0 (b) 2

(c) 3 (d) 1

$(\sqrt{2}+1)(\sqrt{2}-1)$

$= (\sqrt{2})^2 - (1)^2$

$= 2-1$

$= 1$

8. (cosecA – cotA) (1+cosA) =?

SSC CGL 23/08/2021 (Shift- 02)

(a) cosecA (b) cosA

(c) sinA (d) cotA

$$\left(\frac{1}{S}-\frac{C}{S}\right)(1+C)$$

$$= \frac{(1-C)\times(1+C)}{S}$$

$$= \frac{(1-C^2)}{S} = \frac{S^2}{S} = \sin A$$

9. $(\text{cosec}\theta - \cot\theta)^2 =, 0° < \theta < 90°$

SSC CHSL 09/06/2022 (Shift- 03)

(a) $\frac{1-\sin\theta}{1+\cos\theta}$

(b) $\frac{1-\cos\theta}{1+\cos\theta}$

(c) $\frac{1+\cos\theta}{1-\cos\theta}$

(d) $\frac{1+\cos\theta}{1-\sin\theta}$

$\left(\frac{1}{S}-\frac{C}{S}\right)^2 = \frac{(1-C)^2}{S^2} = \frac{(1-C)^2}{1^2-C^2} = \frac{(1-C)(1-C)}{(1-C)(1+C)}$

10. The given expression is equal to:

$$\frac{(1+\tan^2 A)}{\cos ec^2 A.\tan A}$$

$$\frac{\sec^2 A}{\text{cosec}^2 A \times \tan A}$$

$$= \frac{\sin^2 A}{\cos^2 A \times \frac{\sin A}{\cos A}} = \tan A$$

$$\frac{(1+\tan^2 A)}{\cos ec^2 A.\tan A}$$ दी गई अभिव्यक्ति इसके बराबर है:

SSC CHSL 03/08/2023 (Shift-03)

(a) $\sec^2 A$

(b) $\sec A$

(c) $\tan A$ ✓

(d) $\cot A$

TYPE-II

Important Facts Useful for Solving Question on Trigonometry

त्रिकोणमिति के प्रश्नों को हल करने में मदद करने वाले महत्वपूर्ण तथ्य

(1) $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}} = \frac{1-\sin\theta}{\cos\theta} = \sec\theta - \tan\theta$ & $\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta$

(2) $\sqrt{\frac{\text{cosec}\theta-1}{\text{cosec}\theta+1}} = \sec\theta - \tan\theta$ & $\sqrt{\frac{\text{cosec}\theta+1}{\text{cosec}\theta-1}} = \sec\theta + \tan\theta$

(3) $\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \frac{1-\cos\theta}{\sin\theta} = \text{cosec}\theta - \cot\theta$ & $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \text{cosec}\theta + \cot\theta$

(4) $\sqrt{\frac{\sec\theta-1}{\sec\theta+1}} = \text{cosec}\theta - \cot\theta$ & $\sqrt{\frac{\sec\theta+1}{\sec\theta-1}} = \text{cosec}\theta + \cot\theta$

$$* \sqrt{\frac{1+S}{1-S}} = \sqrt{\frac{(1+S)}{(1-S)} \times \frac{(1+S)}{(1+S)}} = \sqrt{\frac{(1+S)^2}{(1-S^2)}} = \frac{(1+S)}{C}$$

$$= \frac{1}{C} + \frac{S}{C}$$

$$= \sec\theta + \tan\theta$$

$$* \sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta$$

$$* \sqrt{\frac{1-\sin\theta}{1+\sin\theta}} = \sec\theta - \tan\theta$$

$$* \sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \operatorname{cosec}\theta + \cot\theta$$

$$* \sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \operatorname{cosec}\theta - \cot\theta$$

$$\frac{S+1-C}{S-1+C} = \frac{\frac{S}{C}+\frac{1}{C}-1}{\frac{S}{C}-\frac{1}{C}+1}$$

(5) $\dfrac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1} = \dfrac{1+\sin\theta}{\cos\theta} = \sec\theta+\tan\theta$

(6) $\dfrac{\tan\theta-\sec\theta+1}{\tan\theta+\sec\theta-1} = \dfrac{1-\sin\theta}{\cos\theta} = \sec\theta-\tan\theta$

(7) $\dfrac{\cot\theta+\operatorname{cosec}\theta-1}{\cot\theta-\operatorname{cosec}\theta+1} = \dfrac{1+\cos\theta}{\sin\theta} = \operatorname{cosec}\theta+\cot\theta$

(8) $\dfrac{\cot\theta-\operatorname{cosec}\theta+1}{\cot\theta+\operatorname{cosec}\theta-1} = \dfrac{1-\cos\theta}{\sin\theta} = \operatorname{cosec}\theta-\cot\theta$

(9) $\dfrac{\sin\theta-\cos\theta+1}{\sin\theta+\cos\theta-1} = \dfrac{1+\sin\theta}{\cos\theta} = \sec\theta+\tan\theta$

(10) $\dfrac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} = \dfrac{1-\sin\theta}{\cos\theta} = \sec\theta-\tan\theta$

(11) $\dfrac{\cos\theta-\sin\theta+1}{\cos\theta+\sin\theta-1} = \dfrac{1+\cos\theta}{\sin\theta} = \operatorname{cosec}\theta+\cot\theta$

(12) $\dfrac{\cos\theta+\sin\theta-1}{\cos\theta-\sin\theta+1} = \dfrac{1-\cos\theta}{\sin\theta} = \operatorname{cosec}\theta-\cot\theta$

11. Find the value of the following expression.

$$\sqrt{\frac{1+\sin\theta}{1-\sin\theta}}$$

SSC CGL 05/12/2022 (Shift-04)

(a) $\sec\theta + \tan\theta$ **(b) $\operatorname{cosec}\theta + \tan\theta$**

(c) $\operatorname{cosec}\theta + \cot\theta$ **(d) $\sec\theta + \cot\theta$**

Trick :

$\sec\theta+\tan\theta$

$\sqrt{\frac{1+\frac{3}{5}}{1-\frac{3}{5}}} = \sqrt{\frac{\frac{8}{5}}{\frac{2}{5}}}^{4} = 2$

11. Find the value of the following expression.

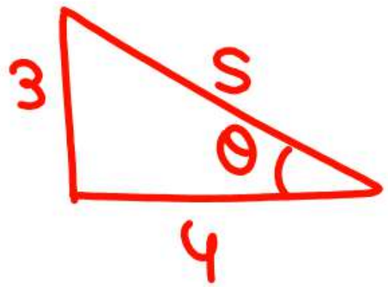
$$\sqrt{\frac{1+\sin\theta}{1-\sin\theta}}$$

SSC CGL 05/12/2022 (Shift-04)

(a) $\sec\theta + \tan\theta$ **(b) $\operatorname{cosec}\theta + \tan\theta$**

(c) $\operatorname{cosec}\theta + \cot\theta$ **(d) $\sec\theta + \cot\theta$**

(a) $\frac{5}{4} + \frac{3}{4} = \frac{8}{4}^{2}$



* $\sqrt{\frac{\operatorname{cosec}\theta+1}{\operatorname{cosec}\theta-1}} = \sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sec\theta+\tan\theta$

* $\sqrt{\frac{\operatorname{cosec}\theta-1}{\operatorname{cosec}\theta+1}} = \sqrt{\frac{1-\sin\theta}{1+\sin\theta}} = \sec\theta-\tan\theta$

* $\sqrt{\frac{\sec\theta+1}{\sec\theta-1}} = \sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \operatorname{cosec}\theta+\cot\theta$

* $\sqrt{\frac{\sec\theta-1}{\sec\theta+1}} = \sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \operatorname{cosec}\theta-\cot\theta$

12. Find the value of $\sqrt{\frac{1-\sin 3\theta}{1+\sin 3\theta}}$

SSC CGL 21/07/2023 (Shift-02)

(a) $\sec 3\theta - \tan 3\theta$ (b) $(\sec 3\theta - \tan 3\theta)^2$

(c) $(\sec 3\theta - \tan 3\theta)^3$ (d) $\sec 3\theta + \tan 3\theta$

Trick: $\sec 3\theta - \tan 3\theta$

13. What is the value of

$$\sqrt{\frac{\text{cosec}A+1}{\text{cosec}A-1}}+\sqrt{\frac{\text{cosec}A-1}{\text{cosec}A+1}}=?$$

SSC CHSL 13/03/2023 (Shift-01)

(a) 2 cosA **(b) secA**

(c) cosA **(d) 2 secA**

$$\sqrt{\frac{1+S}{1-S}}+\sqrt{\frac{1-S}{1+S}}$$

$$= \sec\theta+\tan\theta+\sec\theta-\tan\theta$$

$$= 2\sec\theta$$

14. Simplify the given expression

$$\sqrt{\frac{1+\cos P}{1-\cos P}}$$

SSC CGL 25/07/2023 (Shift-03)
SSC CGL 03/12/2022 (Shift-04)
SSC CHSL 17/08/2023 (Shift-03)

(a) cosecP–cotP **(b) secP–tanP**
(c) secP+tanP **(d) cosecP+cotP** ✓

Trick:

cosec P + cot P

15. If A is an acute angle, then $\sqrt{\frac{1-\cos A}{1+\cos A}}+\sqrt{\frac{1+\cos A}{1-\cos A}}$ is equal to:

SSC CGL 05/12/2022 (Shift-02)

SSC CGL 19/09/2024 (Shift-03)

(a) 2cosecA
(b) 2cosA
(c) 2secA
(d) 2sinA

Trick: $\operatorname{cosec}A-\cot A+\operatorname{cosec}A+\cot A$

$=2\operatorname{cosec}A$

16. The given expression is equivalent to:

$$\sqrt{\frac{\sec\theta+1}{\sec\theta-1}}+\sqrt{\frac{\sec\theta-1}{\sec\theta+1}}$$

SSC CGL 18/09/2024 (Shift-03)

(a) $2\sin\theta$
(b) $2\operatorname{cosec}\theta$
(c) $2\tan\theta$
(d) $2\tan\theta\sec\theta$

$$\sqrt{\frac{1+c}{1-c}}+\sqrt{\frac{1-c}{1+c}}$$

$$= \operatorname{cosec}\theta+\cot\theta+\operatorname{cosec}\theta-\cot\theta$$

$$= 2\operatorname{cosec}\theta$$

$$\frac{\frac{S}{C}-S}{\frac{S}{C}+S}=\frac{S\left(\frac{1}{C}-1\right)}{S\left(\frac{1}{C}+1\right)}$$

$$=\frac{\sec A-1}{\sec A+1}$$

17. What is the value of $\frac{\tan A-\sin A}{\tan A+\sin A}$?

SSC CHSL 09/03/2023 (Shift-02)

(a) 1

(b) $\frac{\sec A+1}{\sec A-1}$

(c) $\frac{\sec A-1}{\sec A+1}$ ✓

(d) secA

$$* \frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1} = \frac{(T+S)-(S^2-T^2)}{T-S+1} = \frac{(S+T)-(S-T)(S+T)}{T-S+1}$$

$$= \frac{(S+T)\left[1-(S-T)\right]}{T-S+1}$$

$$= \frac{(S+T)(1-S+T)}{(1-S+T)}$$

$$= \sec\theta+\tan\theta$$

* $\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1} = \frac{\sin\theta-\cos\theta+1}{\sin\theta+\cos\theta-1} = \sec\theta+\tan\theta$

* $\frac{\tan\theta-\sec\theta+1}{\tan\theta+\sec\theta-1} = \frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} = \sec\theta-\tan\theta$

* $\frac{\cot\theta+\operatorname{cosec}\theta-1}{\cot\theta-\operatorname{cosec}\theta+1} = \frac{\cos\theta-\sin\theta+1}{\cos\theta+\sin\theta-1} = \operatorname{cosec}\theta+\cot\theta$

* $\frac{\cot\theta-\operatorname{cosec}\theta+1}{\cot\theta+\operatorname{cosec}\theta-1} = \frac{\cos\theta+\sin\theta-1}{\cos\theta-\sin\theta+1} = \operatorname{cosec}\theta-\cot\theta$

18. Which of the following is equal to

$$\left[\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}\right]?$$

SSC CGL TIER-II 08/08/2022

(a) $\frac{1+\cos\theta}{\sin\theta}$

(b) $\frac{1+\cot\theta}{\tan\theta}$

(c) $\frac{1+\sin\theta}{\cos\theta}$

(d) $\frac{1+\tan\theta}{\cot\theta}$

$= \frac{\sec\theta+\tan\theta}{\frac{1}{c}+\frac{s}{c}} = \frac{1+s}{c}$

19. What is the value of $\frac{\tan\theta - \sec\theta + 1}{\tan\theta + \sec\theta - 1}$?

SSC CHSL 21/03/2023 (Shift-03)

(a) $\sec\theta + \tan\theta$

(b) $\frac{(1 + \sin\theta)}{\cos\theta}$

(c) $2\sec\theta$

(d) $\frac{\cos\theta}{(1 + \sin\theta)}$

$\sec\theta - \tan\theta$

$= \frac{1}{c} - \frac{s}{c} = \frac{(1-s)}{c} \times \frac{(1+s)}{(1+s)}$

$= \frac{(1-s^2)}{c(1+s)}$

$= \frac{c^2}{c(1+s)} = \frac{c}{1+s}$

$$\frac{\frac{3}{4}-\frac{5}{4}+1}{\frac{3}{4}+\frac{5}{4}-1}=\frac{\frac{3-5+4}{4}}{\frac{3+5-4}{4}}$$

$$=\frac{\frac{2}{4}}{\frac{4}{4}}=\frac{1}{2}$$

19. What is the value of $\frac{\tan\theta-\sec\theta+1}{\tan\theta+\sec\theta-1}$?

SSC CHSL 21/03/2023 (Shift-03)

(a) $\sec\theta+\tan\theta$

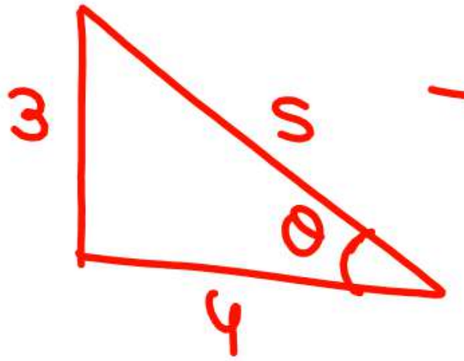
$\frac{5}{4}+\frac{3}{4}=\frac{8}{4}=2$

(b) $\frac{(1+\sin\theta)}{\cos\theta}$

$$\frac{1+\frac{3}{5}}{\frac{4}{5}}=\frac{\frac{8}{5}}{\frac{4}{5}}=2$$

(c) $2\sec\theta$

(d) $\frac{\cos\theta}{(1+\sin\theta)}$



$$\frac{\frac{4}{5}}{1+\frac{3}{5}}=\frac{4}{8}=\frac{1}{2}$$

$\sec^2\theta - \tan^2\theta = 1$

$$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$$

$$\Rightarrow (\sec\theta - \tan\theta) = \frac{1}{(\sec\theta + \tan\theta)}$$

or

$$(\sec\theta + \tan\theta) = \frac{1}{(\sec\theta - \tan\theta)}$$

19. What is the value of $\frac{\tan\theta - \sec\theta + 1}{\tan\theta + \sec\theta - 1}$?

SSC CHSL 21/03/2023 (Shift-03)

(a) $\sec\theta + \tan\theta$

(b) $\frac{(1 + \sin\theta)}{\cos\theta}$

(c) $2\sec\theta$

(d) $\frac{\cos\theta}{(1 + \sin\theta)}$

$$(\sec\theta - \tan\theta) = \frac{1}{\sec\theta + \tan\theta} = \frac{1}{\frac{1}{C} + \frac{S}{C}}$$

$$= \frac{C}{1+S}$$

20. What is the value of $\frac{\cot\theta + \operatorname{cosec}\theta - 1}{\cot\theta - \operatorname{cosec}\theta + 1}$?

SSC CHSL 20/03/2023 (Shift-02)

(a) $2\sec\theta$
(b) $2\operatorname{cosec}\theta$
(c) $2\cot\theta$
(d) $(\operatorname{cosec}\theta + \cot\theta)$

$\operatorname{cosec}\theta + \cot\theta$

21. The value of $\frac{\sin\theta + \cos\theta - 1}{\sin\theta - \cos\theta + 1} \times \sqrt{\frac{1 + \sin\theta}{1 - \sin\theta}}$

SSC CGL 2019 Tier-II (16/10/2020)

(a) 1
(b) – 1
(c) – 2
(d) 2

$$\frac{\tan\theta + 1 - \sec\theta}{\tan\theta - 1 + \sec\theta} \times \sqrt{\frac{1+S}{1-S}}$$

$$= (\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$$

$$= \sec^2\theta - \tan^2\theta$$

$$= 1$$

$\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\operatorname{cosec}^2\theta - 1 = \cot^2\theta$

22. Find the value of

$$\frac{\sin\theta + \cos\theta - 1}{\sin\theta - \cos\theta + 1} \times \frac{\tan^2\theta(\operatorname{cosec}^2\theta - 1)}{\sec\theta - \tan\theta}.$$

(a) 0 (b) -1

(c) 1 (d) $\frac{1}{2}$

$$\left(\frac{\tan\theta + 1 - \sec\theta}{\tan\theta - 1 + \sec\theta}\right) \frac{\tan^2\theta \cot^2\theta}{(\sec\theta - \tan\theta)}$$

$$= (\sec\theta - \tan\theta) \times \frac{1}{(\sec\theta - \tan\theta)}$$

23. $\frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = ?$

(a) $\sec A + \tan A$

(b) $\frac{1 + \sin A}{\cos A}$

(c) $\frac{\sin\frac{A}{2} + \cos\frac{A}{2}}{\cos\frac{A}{2} - \sin\frac{A}{2}}$

(d) All of these

$\frac{\tan\theta - 1 + \sec A}{\tan\theta + 1 - \sec A} = (\sec A + \tan A) = \frac{1+S}{C}$

24. If $\frac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta - 1} = p\sec\theta + q\tan\theta$, where $0 < \theta < \frac{\pi}{2}$, then what is p + q equal to?

CDS I 21/04/2024

(a) 0 (b) 1

(c) 2 (d) 4

$$\frac{\tan\theta - 1 + \sec\theta}{\tan\theta + 1 - \sec\theta} = \frac{p\sec\theta + q\tan\theta}{\sec\theta + \tan\theta}$$

P=1 q=1

P+q=2

$$\frac{\sec 2A + \tan 2A}{1 + \sin 2A} = \frac{\sec 2A + \tan 2A}{\cos 2A}$$

$$= \frac{\sin^2 A + \cos^2 A + 2\sin A \cos A}{\cos^2 A - \sin^2 A}$$

$$= \frac{(C+S)^2}{(C-S)(C+S)} = \frac{C+S}{C-S}$$

25. $\dfrac{\sin 2A - \cos 2A + 1}{\sin 2A + \cos 2A - 1}$

(a) $\sin \dfrac{A}{2}$ **(b)** $\cos \dfrac{A}{2}$

(c) $\dfrac{\sin A + \cos A}{\cos A - \sin A}$ **(d)** $\dfrac{\sin A + \cos A}{\sin A - \cos A}$

$$\frac{\tan 2A - 1 + \sec 2A}{\tan 2A + 1 - \sec 2A} = \sec 2A + \tan 2A$$

$\frac{\tan 2A - 1 + \sec 2A}{\tan 2A + 1 - \sec 2A} = \sec 2A + \tan 2A$

$A = 60^\circ$

$\sec 120^\circ + \tan 120^\circ$

$-2 - \sqrt{3}$

$= -(2+\sqrt{3})$

25. $\left\{\frac{\sin 2A - \cos 2A + 1}{\sin 2A + \cos 2A - 1}\right\}$

(a) $\sin\frac{A}{2} = \sin 30^\circ = \frac{1}{2}$ **(b)** $\cos\frac{A}{2} = \frac{\sqrt{3}}{2}$

(c) $\frac{\sin A + \cos A}{\cos A - \sin A}$ **(d)** $\frac{\sin A + \cos A}{\sin A - \cos A}$

$$\frac{\frac{\sqrt{3}}{2} + \frac{1}{2}}{\frac{1}{2} - \frac{\sqrt{3}}{2}} = \frac{\frac{\sqrt{3}+1}{2}}{\frac{1-\sqrt{3}}{2}}$$

$$= \frac{(\sqrt{3}+1)(1+\sqrt{3})}{(1-\sqrt{3})(1+\sqrt{3})} = \frac{(1+\sqrt{3})^2}{-2} = \frac{1+3+2\sqrt{3}}{-2} = -(2+\sqrt{3})$$

Concept of Value Putting

$\theta = 0°, 30°, 45°, 60°, 90°$

Q $\sin^2\theta + \cos^2\theta = ?$

(a) 0 (b) -1 (c) ~~1~~ (d) 2

#ys

$\theta = 0°$ $\quad 0 + 1 = 1$

$\theta = 30°$ $\quad \frac{1}{4} + \frac{3}{4} = \frac{4}{4} = 1$

$\theta = 45°$ $\quad \frac{1}{2} + \frac{1}{2} = 1$

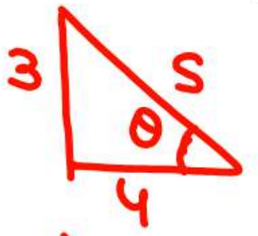
$\theta = 60°$ $\quad \frac{3}{4} + \frac{1}{4} = 1$

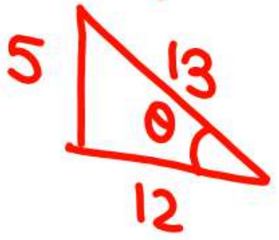
$\theta = 90°$ $\quad 1 + 0 = 1$

Triplets (3,4,5)/(5,12,13) - - -

Q $\sin^2\theta + \cos^2\theta = ?$

(a) 0 (b) -1 (c) 1 (d) 2





$\frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1$

$\frac{25}{169} + \frac{144}{169} = \frac{169}{169} = 1$

26. $\frac{\sin A + \cos A - 1}{\sin A - \cos A + 1} = ?$

(a) $\sec A - \tan A$

(b) $\frac{1 - \sin A}{\cos A}$

(c) $\frac{\cos A}{1 + \sin A}$

(d) All of these

$\frac{\tan A + 1 - \sec A}{\tan A - 1 + \sec A} = \sec A - \tan A$

$= \frac{1 - S}{C}$

27. $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = ?$

(a) $\cot\frac{A}{2}$

(b) $\frac{1+\cos A}{\sin A}$

(c) cosec + cotA

(d) All of the above

$$\frac{\cot A - 1 + \operatorname{cosec} A}{\cot A + 1 - \operatorname{cosec} A} = \operatorname{cosec} A + \cot A$$

$$= \frac{1+\cos A}{\sin A}$$

28. What is $\left(\frac{\cos\theta - \sin\theta + 1}{\cos\theta + \sin\theta - 1}\right)(\cot\theta - \operatorname{cosec}\theta)$ equal to?

CDS II 01/09/2024

(a) -1 (b) 0
(c) 1 (d) 2

$$\left(\frac{\cot\theta - 1 + \operatorname{cosec}\theta}{\cot\theta + 1 - \operatorname{cosec}\theta}\right)(\cot\theta - \operatorname{cosec}\theta)$$

$$= (\cot\theta + \operatorname{cosec}\theta)(\cot\theta - \operatorname{cosec}\theta)$$

$$= \cot^2\theta - \operatorname{cosec}^2\theta$$

$$= -1$$

29. $\dfrac{\cos A + \sin A - 1}{\cos A - \sin A + 1} = ?$

(a) $\tan\dfrac{A}{2}$ (b) $\dfrac{1-\cos A}{\sin A}$

(c) cosec – cotA (d) All of these

$$\frac{\cot A + 1 - \operatorname{cosec}\theta}{\cot A - 1 + \operatorname{cosec}\theta} = \operatorname{cosec}\theta - \cot\theta$$

Result:

$\operatorname{cosec}\theta + \cot\theta = \cot\frac{\theta}{2}$

$\operatorname{cosec}\theta - \cot\theta = \tan\frac{\theta}{2}$

Result:

$\operatorname{cosec}\theta + \cot\theta = \cot\frac{\theta}{2}$

$\operatorname{cosec}\theta - \cot\theta = \tan\frac{\theta}{2}$

Let $\theta = 60^\circ$

L.H.S	R.H.S
$\operatorname{cosec} 60^\circ + \cot 60^\circ$	$\cot 30^\circ$
$= \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}$	$\sqrt{3}$
$= \frac{3}{\sqrt{3}} = \sqrt{3}$	

$\theta = 60^\circ$

$$\frac{1+\sin 60^\circ - \cos 60^\circ}{1+\sin 60^\circ + \cos 60^\circ}$$

$$= \frac{\frac{2}{2}+\frac{\sqrt{3}}{2}-\frac{1}{2}}{\frac{2}{2}+\frac{\sqrt{3}}{2}+\frac{1}{2}} = \frac{\frac{\sqrt{3}+1}{2}}{\frac{\sqrt{3}+3}{2}}$$

$$= \frac{\sqrt{3}+1}{\sqrt{3}(1+\sqrt{3})}$$

$$= \frac{1}{\sqrt{3}}$$

30. **Simplify** $\dfrac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}$.

SSC CGL 24/09/2024 (Shift-03)

(a) $\cos\dfrac{\theta}{2} = \cos 30^\circ = \dfrac{\sqrt{3}}{2}$ **(b)** $\sin\dfrac{\theta}{2}$ $\sin 30^\circ = \dfrac{1}{2}$

(c) $\cot\dfrac{\theta}{2}$ $\cot 30^\circ = \sqrt{3}$ **(d)** $\tan\dfrac{\theta}{2}$ $\tan 30^\circ = \dfrac{1}{\sqrt{3}}$

$\theta = 90^\circ$

$$\frac{1+\sin 90-\cos 90}{1+\sin 90+\cos 90}$$

$$= \frac{1+1-0}{1+1+0} = \frac{2}{2} = 1$$

30. **Simplify** $\dfrac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}$.

SSC CGL 24/09/2024 (Shift-03)

(a) $\cos\dfrac{\theta}{2}$ $\cos 45^\circ = \frac{1}{\sqrt{2}}$

(b) $\sin\dfrac{\theta}{2}$ $\sin 45^\circ = \frac{1}{\sqrt{2}}$

(c) $\cot\dfrac{\theta}{2}$ $\cot 45^\circ = 1$

(d) $\tan\dfrac{\theta}{2}$ $\tan 45^\circ = 1$

$\sin 90^\circ = 1$
$\cos 90^\circ = 0$

$\# \quad \cos 2\theta = 2\cos^2\theta - 1$

$\Rightarrow \quad 1 + \cos 2\theta = 2\cos^2\theta$

$\Rightarrow \quad \frac{1+\cos 2\theta}{2} = \cos^2\theta$

$\therefore \quad \boxed{\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}}}$

$\# \quad \cos 2\theta = 1 - 2\sin^2\theta$

$\Rightarrow \quad 2\sin^2\theta = 1 - \cos 2\theta$

$\Rightarrow \quad \sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\therefore \quad \boxed{\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}}}$

$\therefore$ $\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}}$

$\therefore$ $\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}}$

Ex $\cos 15^\circ = \sqrt{\frac{1+\cos 30^\circ}{2}}$

$= \sqrt{\frac{1+\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2+\sqrt{3}}{4}}$

$= \sqrt{\frac{4+2\sqrt{3}}{8}} = \frac{\sqrt{(\sqrt{3}+1)^2}}{2\sqrt{2}}$

$= \frac{\sqrt{3}+1}{2\sqrt{2}}$ → B, 4

$\sin 15^\circ = \sqrt{\frac{1-\cos 30^\circ}{2}}$

$= \sqrt{\frac{1-\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2-\sqrt{3}}{4}}$

$= \frac{\sqrt{3}-1}{2\sqrt{2}}$ P, 4

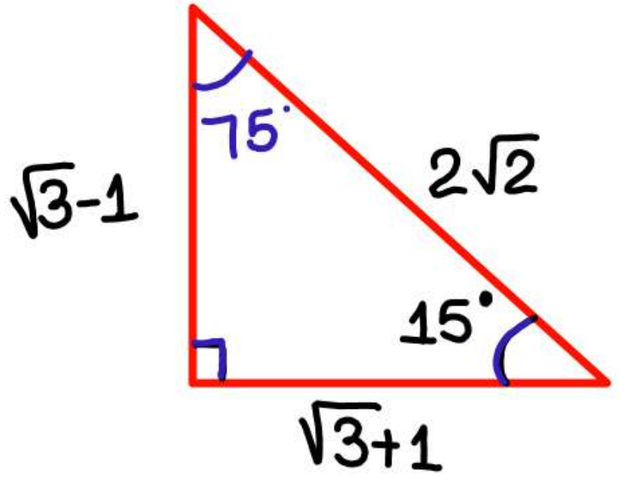
$\therefore$ $\boxed{\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}}}$ $\therefore$ $\boxed{\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}}}$

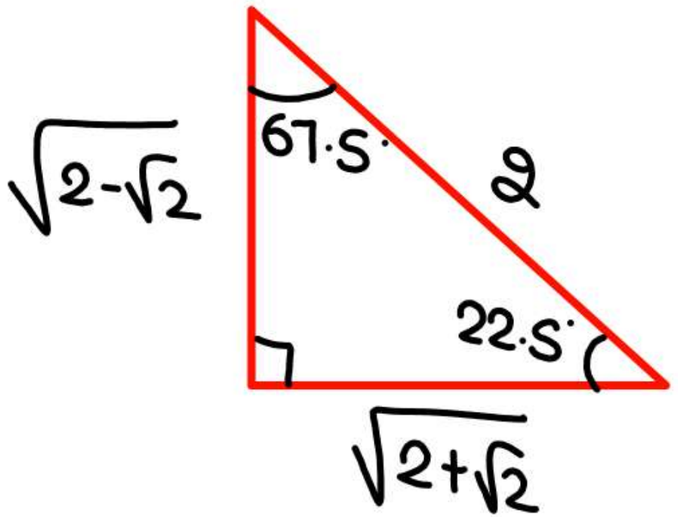
Ex $\cos 22.5^\circ = \sqrt{\frac{1+\cos 45^\circ}{2}}$

$= \sqrt{\frac{1+\frac{1}{\sqrt{2}}}{2}} = \sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$

$= \sqrt{\frac{2+\sqrt{2}}{4}} = \frac{\sqrt{2+\sqrt{2}}}{2}$

$\sin 22.5^\circ = \frac{\sqrt{2-\sqrt{2}}}{2}$


75°
$\sqrt{3}-1$
$2\sqrt{2}$
15°
$\sqrt{3}+1$


67.5°
$\sqrt{2-\sqrt{2}}$
2
22.5°
$\sqrt{2+\sqrt{2}}$

M-1 (Basic)

31. What is the value of $\tan\frac{\theta}{2}$?

SSC CGL 26/07/2023 (Shift-04)

(a) $\frac{\cos\theta}{1-\sin\theta}$

(b) $\frac{\sin\theta}{1-\cos\theta}$

(c) $\frac{\cos\theta}{1-\cos\theta}$

(d) $\frac{\sin\theta}{1+\cos\theta}$

$$\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}} = \sqrt{\frac{\frac{1-\cos\theta}{2}}{\frac{1+\cos\theta}{2}}} = \sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$$

$$= \sqrt{\frac{(1-c)}{(1+c)} \times \frac{(1+c)}{(1+c)}}$$

$$= \sqrt{\frac{(1-c^2)}{(1+c)^2}} = \frac{s}{(1+c)}$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}}$$

$$\cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}}$$

Value Putting

$\theta = 60^\circ$

31. What is the value of $\tan\frac{\theta}{2}$? $= \frac{1}{\sqrt{3}}$

SSC CGL 26/07/2023 (Shift-04)

(a) $\frac{\cos\theta}{1-\sin\theta}$ $\frac{1}{2}$ / $1-\frac{\sqrt{3}}{2}$

(b) $\frac{\sin\theta}{1-\cos\theta}$

(c) $\frac{\cos\theta}{1-\cos\theta}$

(d) $\frac{\sin\theta}{1+\cos\theta}$ $\frac{\frac{\sqrt{3}}{2}}{1+\frac{1}{2}}$

$= \frac{\frac{\sqrt{3}}{2}}{\frac{3}{2}} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$

32. If $\tan\frac{A}{2} = x$, then find x.

SSC CGL 18/07/2023 (Shift-01)

(a) $\frac{\sqrt{1+\cos A}}{\sqrt{1-\cos A}}$

(b) $\frac{\sqrt{1-\sin A}}{\sqrt{1+\cos A}}$

(c) $\frac{\sqrt{1-\cos A}}{\sqrt{1+\cos A}}$

(d) $\frac{\sqrt{\cos A-1}}{\sqrt{1+\cos A}}$

$$\tan\frac{A}{2} = \frac{\sin\frac{A}{2}}{\cos\frac{A}{2}}$$

$$= \frac{\sqrt{\frac{1-\cos A}{2}}}{\sqrt{\frac{1+\cos A}{2}}}$$

33. The value of $1+\sqrt{\frac{\cot\theta+\cos\theta}{\cot\theta-\cos\theta}}$, if $0° < \theta < 90°$, is equal to:

SSC CGL 12/04/2022 (Shift-02)

(a) $1 - \sec\theta + \tan\theta$
(b) $1 - \sec\theta - \tan\theta$
(c) $1 + \sec\theta - \tan\theta$
(d) $1 + \sec\theta + \tan\theta$

$$1+\sqrt{\frac{\frac{c}{s}+c}{\frac{c}{s}-c}} = 1+\sqrt{\frac{c\left(\frac{1}{s}+1\right)}{c\left(\frac{1}{s}-1\right)}} = 1+\sqrt{\frac{1+s}{1-s}}$$

$$= 1+\sec\theta+\tan\theta$$

$$\frac{C}{1+\frac{S}{C}} - \frac{S}{1+\frac{C}{S}}$$

$$= \frac{C^2}{C+S} - \frac{S^2}{S+C}$$

$$= \frac{(C^2-S^2)}{C+S} = \frac{((-S)((C+S)}{((+S)}$$

34. **Simplify:** $\frac{\cos A}{1+\tan A} - \frac{\sin A}{1+\cot A}$

SSC CGL 03/12/2022 (Shift-01)

(a) tanA (b) cosA – sinA

(c) cosA. sinA (d) cosA + sinA

$\theta = 45^\circ$

34. Simplify: $\dfrac{\cos A}{1+\tan A} - \dfrac{\sin A}{1+\cot A}$

SSC CGL 03/12/2022 (Shift-01)

(a) tanA $\quad 1$

(b) cosA – sinA $\quad \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}$ $\quad 0$

(c) cosA. sinA $\quad \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{2}$

(d) cosA + sinA $\quad \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} \sqrt{2}$

$$= \frac{\frac{1}{\sqrt{2}}}{1+1} - \frac{\frac{1}{\sqrt{2}}}{1+1}$$

$$= \frac{\frac{1}{\sqrt{2}}}{2} - \frac{\frac{1}{\sqrt{2}}}{2}$$

$$\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$$

θ = 45°

S = C

T = C

S = C

34. Simplify: $\frac{\cos A}{1+\tan A} - \frac{\sin A}{1+\cot A}$

SSC CGL 03/12/2022 (Shift-01)

(a) tanA

(b) cosA – sinA

(c) cosA. sinA

(d) cosA + sinA

$\frac{S}{1+C} - \frac{S}{1+C}$

= 0

$$\frac{\frac{4}{5}}{1+\frac{3}{4}} - \frac{\frac{3}{5}}{1+\frac{4}{3}}$$

$$= \left(\frac{\frac{4}{5}}{\frac{7}{4}}\right) - \left(\frac{\frac{3}{5}}{\frac{7}{3}}\right)$$

$$= \frac{16}{35} - \frac{9}{35} = \frac{7}{35} = \frac{1}{5}$$

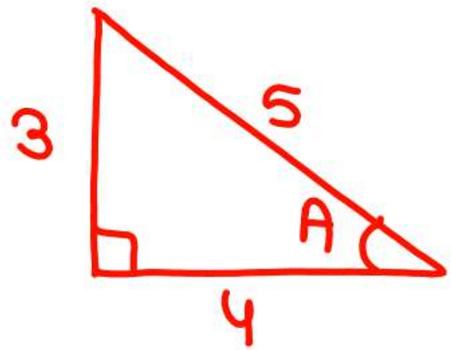
34. Simplify: $\dfrac{\cos A}{1+\tan A} - \dfrac{\sin A}{1+\cot A}$

SSC CGL 03/12/2022 (Shift-01)

(a) $\tan A$ **(b)** $\cos A - \sin A$ ✓

(c) $\cos A . \sin A$ **(d)** $\cos A + \sin A$

$\frac{4}{5} - \frac{3}{5} = \frac{1}{5}$



M-1(Basic)

35. **Simplify:** $\frac{\cos A}{1-\tan A}+\frac{\sin A}{1-\cot A}$

SSC CGL 13/12/2022 (Shift-02)

(a) tanA – cotA (b) tanA + cotA

(c) sinA – cosA (d) sinA + cosA

$$\frac{C}{1-\frac{S}{C}}+\frac{S}{1-\frac{C}{S}}$$

$$=\frac{C^2}{C-S}+\frac{S^2}{S-C}$$

$$=\frac{C^2}{C-S}-\frac{S^2}{C-S}$$

$$=\frac{C^2-S^2}{C-S}=\frac{(C-S)(C+S)}{(C-S)}$$

$$\frac{\frac{4}{5}}{1-\frac{3}{4}} + \frac{\frac{3}{5}}{1-\frac{4}{3}}$$

$$= \left(\frac{\frac{4}{5}}{\frac{1}{4}}\right) + \left(\frac{\frac{3}{5}}{-\frac{1}{3}}\right)$$

$$= \frac{16}{5} - \frac{9}{5}$$

$$= \frac{7}{5}$$

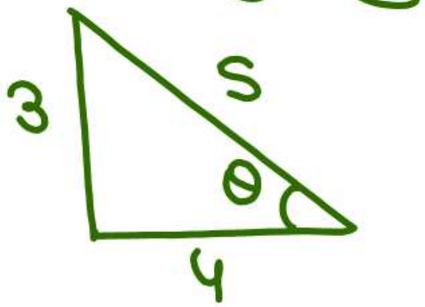
35. **Simplify:** $\dfrac{\cos A}{1-\tan A} + \dfrac{\sin A}{1-\cot A}$

SSC CGL 13/12/2022 (Shift-02)

(a) tanA – cotA (b) tanA + cotA

(c) sinA – cosA (d) sinA + cosA

$\frac{3}{5} + \frac{4}{5} = \frac{7}{5}$



$\theta = 45^\circ$

$\theta = 135^\circ$

- $\sin 135^\circ = \sin(90+45) = \cos 45^\circ = \frac{1}{\sqrt{2}}$
- $\cos 135 = \cos(90+45^\circ) = -\sin 45 = \frac{-1}{\sqrt{2}}$
- $\tan 135 = -1$

  or

  $\cot 135^\circ = -1$

35. Simplify: $\frac{\cos A}{1-\tan A}+\frac{\sin A}{1-\cot A}$

SSC CGL 13/12/2022 (Shift-02)

(a) tanA – cotA (b) tanA + cotA

(c) sinA – cosA (d) sinA + cosA

$\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$

$\frac{-\frac{1}{\sqrt{2}}}{2}+\frac{\frac{1}{\sqrt{2}}}{2}$

$= 0$

$A = 135^\circ$

$$\frac{-\frac{1}{\sqrt{2}}}{2} + \frac{\frac{1}{\sqrt{2}}}{2} - \frac{1}{\sqrt{2}}$$

36. Simplify the following expression:

$$\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} - \sin A$$

$(1 - (-1))$

SSC CGL 19/04/2022 (Shift-03)

(a) 1 + sinA

(b) (1 + sinA) cosA

(c) cosA

(d) 1 + cosA

$-\frac{1}{\sqrt{2}}$

37. Solve the following:

$(\tan\theta + \sec\theta + 1)(\cot\theta - \operatorname{cosec}\theta + 1) = ?$

SSC CPO 29/06/2024 (Shift-01)

(a) 2 **(b) 1**

(c) 0 **(d) -1**

$\theta = 45^\circ$

$(1+\sqrt{2}+1)(1-\sqrt{2}+1)$

$= (2+\sqrt{2})(2-\sqrt{2})$

$= (2^2-2)$

$= 2$

38. What is the value of

$(1 + \tan\theta - \sec\theta)(1 + \cot\theta + \operatorname{cosec}\theta)$?

SSC CHSL 09/03/2023 (Shift-03)

(a) -1 (b) 0

(c) 1 (d) 2

A = 45°

39. **The value of**

(1 + cot A – cosec A)(1 + tanA + sec A)–1 is:

SSC CPO 09/11/2022 (Shift-02)

(a) 1 **(b) 2**

(c) 3 **(d) 0**

$(1+1-\sqrt{2})(1+1+\sqrt{2})-1$

$= (2-\sqrt{2})(2+\sqrt{2})-1$

$= (4-2)-1$

$= 2-1$

$= 1$

40. The value of

$(\text{cosec}A + \cot A + 1)(\text{cosec}A - \cot A + 1) - 2\text{cosec}A$ is

SSC CGL 09/03/2020 (Shift-02)

(a) 4cosecA (b) 2

(c) 2cosecA (d) 0

$A = 45^\circ$

$(\sqrt{2}+2)(\sqrt{2}) - 2\sqrt{2}$

$= 2 + 2\sqrt{2} - 2\sqrt{2}$

A=45°

41. Find the value of

$(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 3(\sin^2 A + \cos^2 A)$.

SSC CPO 27/06/2024 (Shift-02)

(a) –1 **(b) 1**

(c) 2 **(d) –2**

$(2-\sqrt{2})(2+\sqrt{2})-3(1)$

$= (4-2)-3$

$= 2-3$

$= -1$

$\frac{\sin\theta}{1+\cos\theta}$

Soln $\frac{S}{(1+C)} \times \frac{(1-C)}{(1-C)}$

$= \frac{S(1-C)}{1^2-C^2}$

$= \frac{S(1-C)}{S^2}$

$= \frac{1-\cos\theta}{\sin\theta}$

$\frac{\sin\theta}{1-\cos\theta}$

$= \frac{S}{(1-C)} \times \frac{(1+C)}{(1+C)}$

$= \frac{S(1+C)}{1^2-C^2}$

$= \frac{S(1+C)}{S^2}$

$= \frac{1+\cos\theta}{\sin\theta}$

$\frac{\cos\theta}{1+\sin\theta}$

$= \frac{C}{(1+S)} \frac{(1-S)}{(1-S)}$

$= \frac{C(1-S)}{1-S^2}$

$= \frac{C(1-S)}{C^2}$

$= \frac{1-\sin\theta}{\cos\theta}$

$\frac{\cos\theta}{1-\sin\theta}$

$= \frac{C}{(1-S)} \times \frac{(1+S)}{(1+S)}$

$= \frac{C(1+S)}{1-S^2}$

$= \frac{C(1+S)}{C^2}$

$= \frac{1+\sin\theta}{\cos\theta}$

$$\# \frac{\sin\theta}{1+\cos\theta} = \frac{1-\cos\theta}{\sin\theta}$$

$$\# \frac{\sin\theta}{1-\cos\theta} = \frac{1+\cos\theta}{\sin\theta}$$

$$\# \frac{\cos\theta}{1+\sin\theta} = \frac{1-\sin\theta}{\cos\theta}$$

$$\# \frac{\cos\theta}{1-\sin\theta} = \frac{1+\sin\theta}{\cos\theta}$$

42. If A is an acute angle, which of the following is equal to $\frac{\sin A}{1+\cos A}$?

SSC CGL 09/09/2024 (Shift-03)

(a) $\frac{1-\cos A}{\sin A}$

(b) $\frac{1+\cos A}{\sin A}$

(c) $\frac{1-\sin A}{\cos A}$

(d) $\frac{1+\sin A}{\cos A}$

$\frac{1-\cos A}{\sin A}$

43. If $p = \frac{\sin A}{1+\cos A}$ then $\frac{\sin A}{1-\cos A}$ is equal to:

SSC CGL 26/09/2024 (Shift-02)

(a) $\frac{1}{p} = \frac{1+\cos A}{\sin A}$

(b) $\frac{1}{1-p}$

(c) $\frac{1}{p+1}$

(d) $p - 1$

44. Simplify: $\frac{\sin A}{1+\cos A} + \cot A$, where A is an acute angle.

SSC CGL 18/09/2024 (Shift-02)

(a) SinA (b) CosA

(c) sinA + cosA (d) cosecA

$$\frac{1-\cos A}{\sin A} + \frac{\cos A}{\sin A}$$

$$= \frac{1-\cos A+\cos A}{\sin A}$$

$$= \operatorname{cosec} A$$

Basic

$$\sqrt{\frac{1-\frac{S}{C}}{1+\frac{S}{C}}} = \sqrt{\frac{(C-S)}{(C+S)} \times \frac{(C-S)}{(C-S)}}$$

$$= \sqrt{\frac{(C-S)^2}{C^2-S^2}}$$

$$= \sqrt{\frac{C^2+S^2-2\times S\times C}{C^2-S^2}}$$

$$= \sqrt{\frac{1-\sin 2A}{\cos 2A}}$$

45. Find the value of $\sqrt{\dfrac{1-\tan A}{1+\tan A}}$

SSC CPO 04/10/2023 (Shift-02)

(a) $\sqrt{\dfrac{1+\sin A}{\cos A}}$

(b) $\sqrt{\dfrac{1+\sin 2A}{\cos 2A}}$

(c) $\sqrt{\dfrac{1-\sin A}{\cos A}}$

(d) $\sqrt{\dfrac{1-\sin 2A}{\cos 2A}}$

- $\sin 2\theta = 2\sin\theta\cos\theta$
- $\cos 2\theta = \cos^2\theta - \sin^2\theta$

TYPE-III

* $\tan\theta + \frac{1}{\tan\theta} = 2$

or

$\tan\theta + \cot\theta = 2$

$\tan\theta = 1$

$\theta = 45^\circ$

* $\sin\theta + \frac{1}{\sin\theta} = 2$

or

$\sin\theta + \operatorname{cosec}\theta = 2$

$\sin\theta = 1$

$\theta = 90^\circ$

* $\cos\theta + \frac{1}{\cos\theta} = 2$

or

$\cos\theta + \sec\theta = 2$

$\cos\theta = 1$

$\theta = 0^\circ$

1 + 1

46. If $\tan\theta + \cot\theta = 2$ and θ is an acute angle, then the value of θ is:

SSC CGL 19/09/2024 (Shift-03)

(a) 30°
(b) 15°
(c) 60°
(d) 45°

A = 45°

47. If A is an acute angle and $\tan A + \cot A = 2$, find the value of $7\tan^3 A - 6\cot^3 A + 8\sec^2 A$.

SSC CGL 26/09/2024 (Shift-01)

(a) 6
(b) 17
(c) 16
(d) 7

$7 \times 1 - 6 \times 1 + 8 \times 2$

$= 17$

48. If θ is an angle and $\sin\theta + \operatorname{cosec}\theta = 2$, then the value of $\sin^5\theta + \operatorname{cosec}^5\theta$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) 2
(b) 10
(c) 4
(d) 5

$\sin\theta + \operatorname{cosec}\theta = 2$

$\sin\theta = 1$

$1 + 1$
$= 2$

$\cos\theta + \sec\theta = 2$

$\cos\theta = 1$

$\theta = 0^\circ$

49. If $\cos\theta + \sec\theta = 2$, then the value of $\sin^6\theta + \cos^6\theta$ is:

0 + 1

SSC CPO 10/11/2022 (Shift-03)

(a) $\frac{1}{3}$

(b) 0

(c) 1

(d) $\frac{1}{2}$

Algebra. $x+\frac{1}{x}=n$ $\quad x^2+\frac{1}{x^2}=n^2-2$

Trigo $\quad \sin\theta+\frac{1}{\sin\theta}=n$ $\quad \sin^2\theta+\frac{1}{\sin^2\theta}=n^2-2$

Ex: $\sin\theta+\operatorname{cosec}\theta=5$ $\quad \sin^2\theta+\operatorname{cosec}^2\theta=5^2-2=23$

$\cos\theta+\frac{1}{\cos\theta}=n$, $\cos^2\theta+\frac{1}{\cos^2\theta}=n^2-2$

$\tan\theta+\frac{1}{\tan\theta}=n$ $\quad \tan^2\theta+\frac{1}{\tan^2\theta}=n^2-2$

- $x+\frac{1}{x}=n$ $\quad x^3+\frac{1}{x^3}=n^3-3n$
- $\sin\theta+\frac{1}{\sin\theta}=n$ $\quad \sin^3\theta+\frac{1}{\sin^3\theta}=n^3-3n$
- $\cos\theta+\frac{1}{\cos\theta}=n$ $\quad \cos^3\theta+\frac{1}{\cos^3\theta}=n^3-3n$

50. If $\tan\theta + \cot\theta = 6$, then the value of $\tan^2\theta + \cot^2\theta$ is ____ .

SSC CPO 28/06/2024 (Shift-02)

(a) 24
(b) 34
(c) 20
(d) 22

$\tan\theta + \cot\theta = 6$

$\tan^2\theta + \cot^2\theta = 6^2 - 2$

$= 34$

50. If $\tan\theta + \cot\theta = 6$, then the value of $\tan^2\theta + \cot^2\theta$ is ____ .

SSC CPO 28/06/2024 (Shift-02)

(a) 24+2=26
(b) 34+2=36
(c) 20+2=22
(d) 22+2=24

51. If $\sec\theta + \cos\theta = 32$, then $\sec^2\theta + \cos^2\theta$ is _____.

SSC CGL 14/07/2023 (Shift-04)

(a) 1000
(b) 1022
(c) 1024
(d) 1020

Ans $= (32)^2 - 2$

$= 1024 - 2$

$= 1022$

52. If $\sin\theta + \operatorname{cosec}\theta = \sqrt{5}$, then the value of $\sin^3\theta + \operatorname{cosec}^3\theta$ is:

SSC Phase-XII 26/06/2024 (Shift-03)

(a) $\sqrt{5}$

(b) 0

(c) $\frac{1}{\sqrt{5}}$

(d) $2\sqrt{5}$

ans $= n^3 - 3n$

$= (\sqrt{5})^3 - 3 \times \sqrt{5}$

$= 5\sqrt{5} - 3\sqrt{5}$

$= 2\sqrt{5}$

$x-\frac{1}{x}=n$, $x+\frac{1}{x}=\sqrt{n^2+4}$

$\sin\theta-\frac{1}{\sin\theta}=n$, $\sin\theta+\frac{1}{\sin\theta}=\sqrt{n^2+4}$

or

$\sin\theta-\operatorname{cosec}\theta=n$, $\sin\theta+\operatorname{cosec}\theta=\sqrt{n^2+4}$

$\cos\theta$, $\tan\theta$

$x+\frac{1}{x}=n$, $x-\frac{1}{x}=\sqrt{n^2-4}$

53. If $\sin\theta - \operatorname{cosec}\theta = 6$, then what is the value of $\sin\theta + \operatorname{cosec}\theta$?

SSC CHSL 14/03/2023 (Shift-02)

(a) 6 **(b) $\sqrt{40}$**

(c) $\sqrt{34}$ **(d) $\sqrt{38}$**

$\sin\theta - \operatorname{cosec}\theta = 6$, $\sin\theta + \operatorname{cosec}\theta = \sqrt{n^2+4}$

$= \sqrt{40}$

54. If $3x = \sec A$ and $\frac{3}{x} = \tan A$ then $9\left(x^2 - \frac{1}{x^2}\right)$ is ——.

SSC CGL 27/07/2023 (Shift-02)

(a) 3 (b) 9

(c) 1 (d) $\frac{1}{9}$

$9x^2 - \frac{9}{x^2}$

$= (3x)^2 - \left(\frac{3}{x}\right)^2$

$= \sec^2 A - \tan^2 A$

$= 1$

TYPE-IV

Special Concept

$\boxed{\sec^2\theta - \tan^2\theta = 1}$ or $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\Rightarrow (\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$

$(\sec\theta - \tan\theta) = \dfrac{1}{(\sec\theta + \tan\theta)}$

If $\sec\theta + \tan\theta = p$

$\sec\theta - \tan\theta = \dfrac{1}{p}$

$(\sec\theta + \tan\theta) = \dfrac{1}{\sec\theta - \tan\theta}$

If $\sec\theta - \tan\theta = p$

Then $\sec\theta + \tan\theta = \dfrac{1}{p}$

Q1. $\sec\theta + \tan\theta = 5$

$\sec\theta - \tan\theta = \frac{1}{5}$

Q2. $\operatorname{cosec}\theta + \cot\theta = 7$

$\operatorname{cosec}\theta - \cot\theta = \frac{1}{7}$

$\sec\theta \pm \tan\theta = P$

$+ \quad \sec\theta \mp \tan\theta = \frac{1}{P}$

$2\sec\theta = P + \frac{1}{P}$

$\therefore 2\sec\theta = \frac{P^2+1}{P}$

$\sec\theta = \frac{P^2+1}{2P}$

$\sec\theta + \tan\theta = P$

$\therefore (-) \quad \sec\theta - \tan\theta = \frac{1}{P}$

$2\tan\theta = P - \frac{1}{P}$

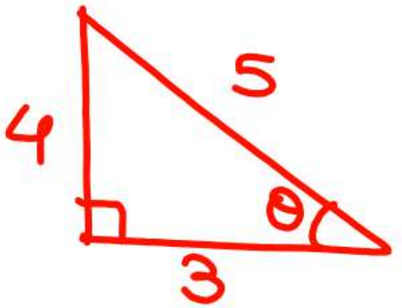
$2\tan\theta = \frac{P^2-1}{P}$

$\tan\theta = \frac{P^2-1}{2P}$

Q1. $\sec\theta + \tan\theta = 3$

$\sec\theta = ?$

Soln $\sec\theta = \frac{p^2+1}{2p} = \frac{9+1}{6} = \frac{10}{6} = \frac{5}{3}$ → H, B



✓ $\sin\theta = \frac{4}{5}$

✓ $\tan\theta = \frac{4}{3}$

55. If $\sec\theta + \tan\theta = 5$, then what is the value of $\sec\theta - \tan\theta$? $= \frac{1}{5}$

SSC CHSL 09/03/2023 (Shift-01)

(a) $\frac{1}{5}$

(b) $\frac{1}{3}$

(c) $\frac{1}{6}$

(d) $\frac{1}{2}$

$\sec\theta + \tan\theta = p$

$\sec\theta - \tan\theta = \frac{1}{p}$

56. If the value of cosecA + cotA = m, then the value of cosecA – cotA is:

SSC CPO 10/11/2022 (Shift-02)

(a) $\frac{1}{m}$

(b) m

(c) $\sqrt{m}$

(d) m^2

$\sec\theta = \frac{x^2+1}{2x}$ → H, B

$\sin\theta = \frac{x^2-1}{x^2+1}$

57. If $\sec\theta + \tan\theta = x$, then find $\sin\theta$.

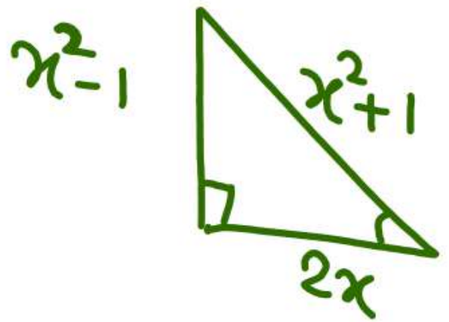
SSC CGL 25/09/2024 (Shift-03)

(a) $\frac{x^2+1}{1-x^2}$

(b) $\frac{x^2-1}{1+2x^2}$

(c) $\frac{x^2-1}{1+x^2}$

(d) $\frac{1-x^2}{1+x^2}$



58. If $\sec\theta + \tan\theta = 5$, $(\theta \neq 0)$ then $\sec\theta$ is equal to:

SSC CGL 14/07/2023 (Shift-04)

(a) $\left(5+\frac{1}{5}\right)$

(b) $\frac{1}{2}\left(3+\frac{1}{3}\right)$

(c) $\frac{1}{2}\left(5+\frac{1}{5}\right)$

(d) $\left(3+\frac{1}{3}\right)$

$\sec\theta + \tan\theta = 5$

(+) $\sec\theta - \tan\theta = \frac{1}{5}$

$2\sec\theta = \left(5+\frac{1}{5}\right)$

$\sec\theta = \frac{1}{2}\left(5+\frac{1}{5}\right)$

$\sec\theta = \frac{p^2+1}{2p} = \frac{3+1}{2\sqrt{3}}$

$= \frac{4}{2\sqrt{3}}$ → H / B

$\text{ans} = \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1+\sqrt{3}}{2}$

59. **If $\sec\theta + \tan\theta = \sqrt{3}$ then find the positive value of $\sin\theta + \cos\theta$.**

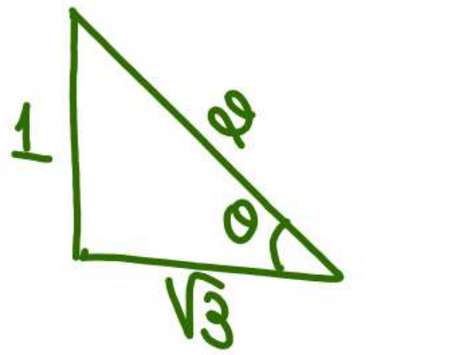
SSC CHSL 04/08/2023 (Shift-03)

(a) $\frac{\sqrt{3}}{2}$

(b) $\frac{2}{\sqrt{3}}$

(c) $\frac{2}{(1+\sqrt{3})}$

(d) $\frac{(1+\sqrt{3})}{2}$



60. If $\sec^2\theta + \tan^2\theta = \frac{25}{18}$, then the value of $\sec^4\theta - \tan^4\theta$ is:

SSC CGL 14/07/2023 (Shift-03)

(a) $\frac{18}{25}$

(b) $\frac{25}{12}$

(c) $\frac{25}{9}$

(d) $\frac{25}{18}$

$(\sec^4\theta - \tan^4\theta) = (\sec^2\theta - \tan^2\theta)(\sec^2\theta + \tan^2\theta)$

$= 1 \times \frac{25}{18}$

$= \frac{25}{18}$

$(a^2 - b^2) = (a-b)(a+b)$

$(a^4 - b^4) = (a^2 - b^2)(a^2 + b^2)$

$(a^6 - b^6) = (a^3 - b^3)(a^3 + b^3)$

61. If $\text{cosec}\theta + \cot\theta = \frac{3}{2}$, then what is the value of $\text{cosec}\theta$? $= \frac{p^2+1}{2p}$

SSC CGL 24/07/2023 (Shift-03)

(a) $\frac{13}{12}$

(b) $\frac{3}{2}$

(c) $\frac{11}{12}$

(d) $\frac{9}{13}$

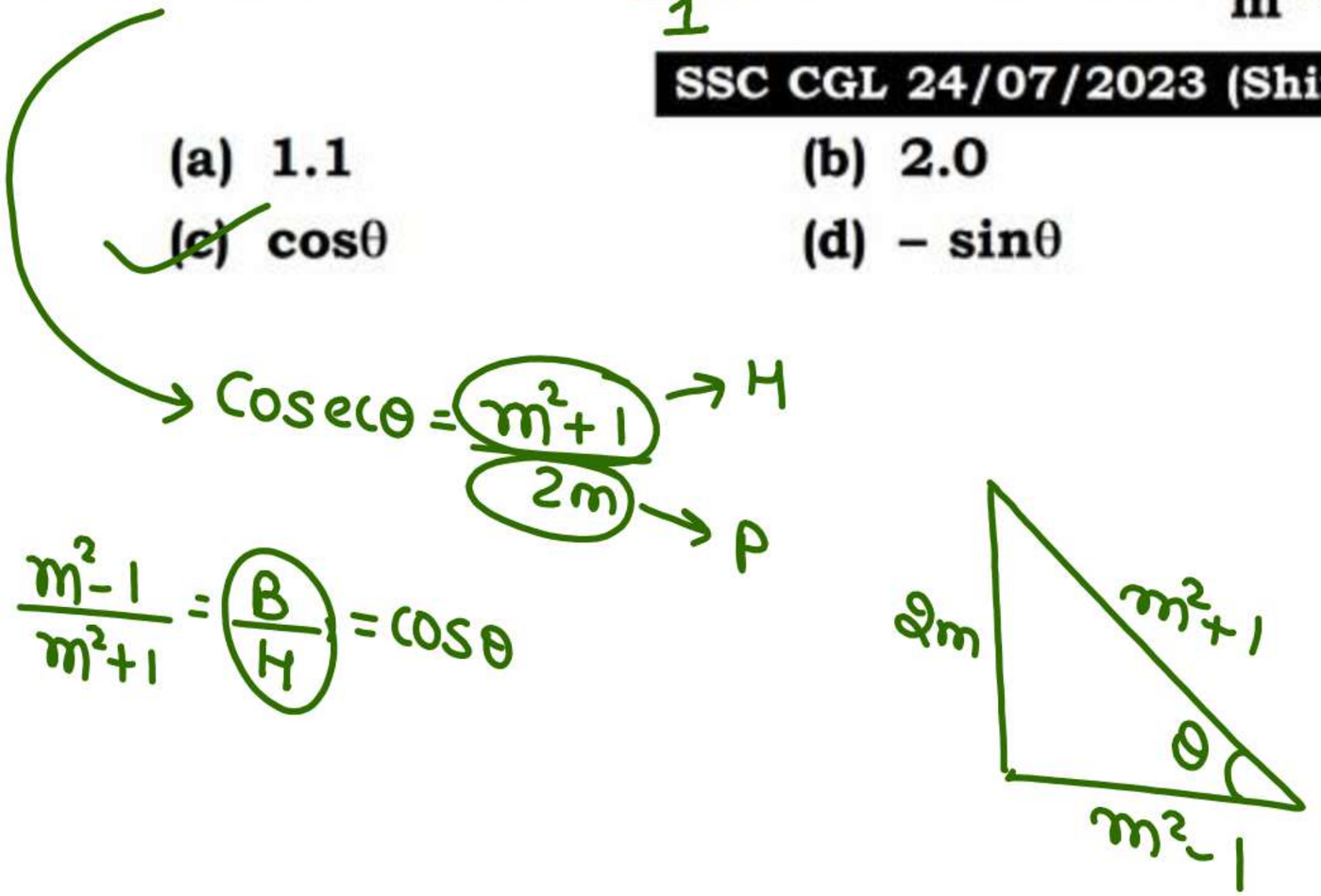
ans $= \frac{\frac{9}{4}+1}{\frac{3}{2}} = \frac{\frac{13}{4}}{3} = \frac{13}{12}$

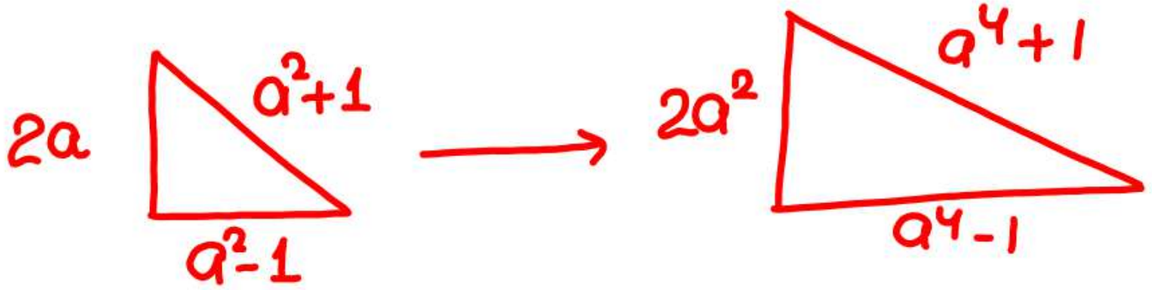
62. If $\text{cosec}\theta + \cot\theta = \frac{m}{1}$, find the value of $\frac{m^2 - 1}{m^2 + 1}$.

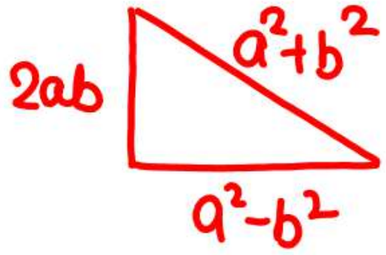
SSC CGL 24/07/2023 (Shift-02)

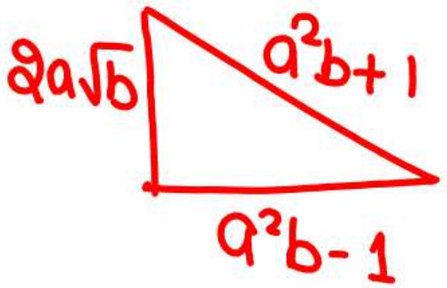
(a) 1.1 (b) 2.0

(c) $\cos\theta$ (d) $-\sin\theta$




2a
a^2+1
a^2-1
$2a^2$
a^4+1
a^4-1


2ab
a^2+b^2
a^2-b^2


$2a\sqrt{b}$
a^2b+1
a^2b-1

$\sec\theta + \tan\theta = m$ or $\sec\theta - \tan\theta = m$

$$\sec\theta = \frac{m^2+1}{2m}$$

$\operatorname{cosec}\theta + \cot\theta = m$ or $\operatorname{cosec}\theta - \cot\theta = m$

$$\operatorname{cosec}\theta = \frac{m^2+1}{2m}$$

63. If $\text{cosec}A + \cot A = a\sqrt{b}$, then find the value of $\dfrac{(a^2b-1)}{(a^2b+1)}$.

$= \dfrac{B}{H}$

SSC CHSL 17/08/2023 (Shift-02)

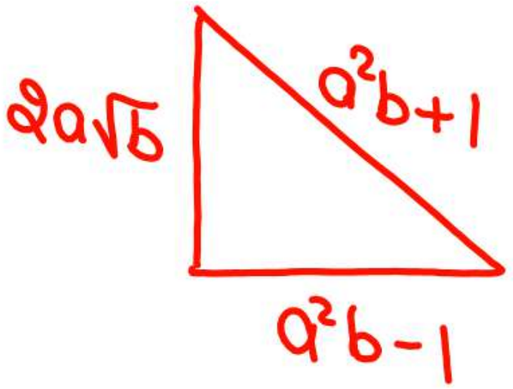
(a) cosA

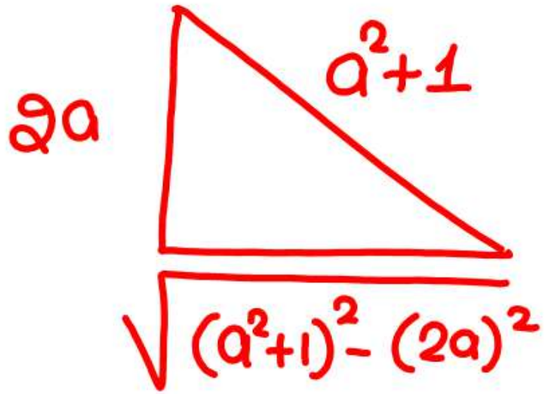
(b) tanA

(c) $\dfrac{1}{\sin A}$

(d) $\dfrac{1}{\cot A}$

$\text{Cosec}\theta = \dfrac{a^2b+1}{2a\sqrt{b}}$ → H, → P



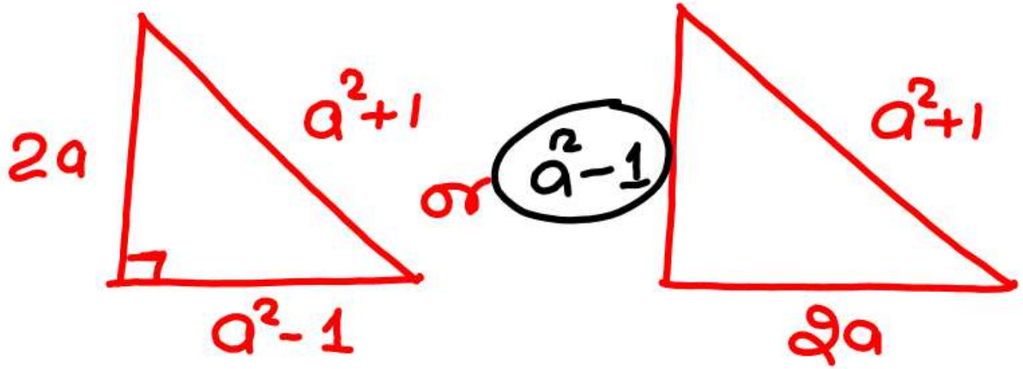


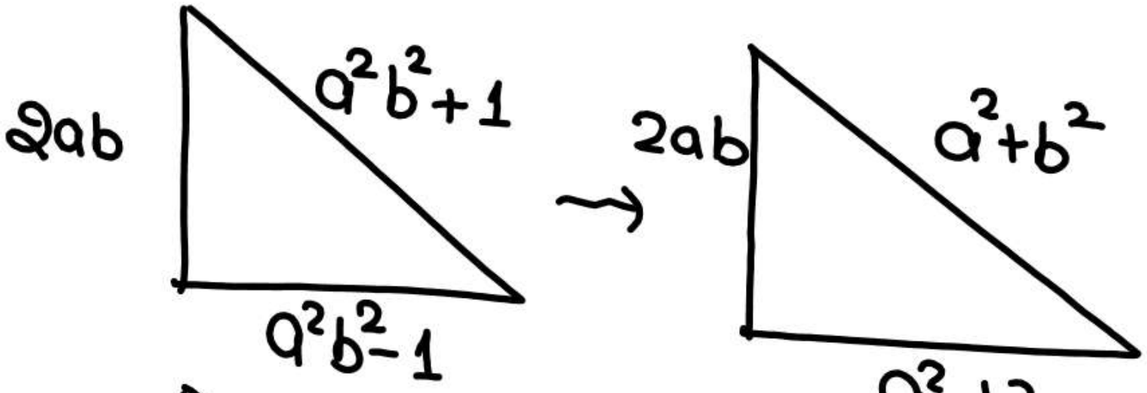
$$= \sqrt{a^4+1+2a^2-4a^2}$$

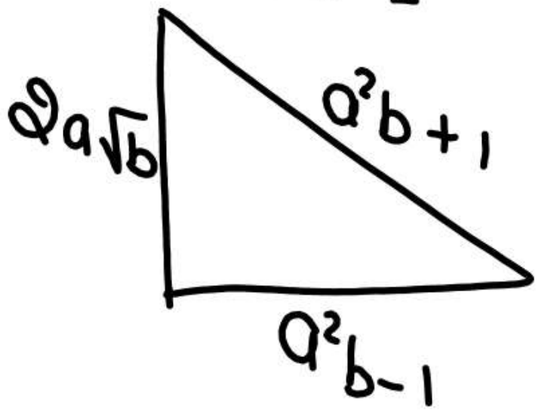
$$= \sqrt{a^4+1-2a^2}$$

$$= \sqrt{(a^2-1)^2}$$

$$= (a^2-1)$$







64. If $\sec x + \tan x = 5$ and $\operatorname{cosec} y - \cot y = \frac{1}{3}$, then find the value of $(\sec x + \operatorname{cosec} y) - (\tan x - \cot y)$.

SSC CHSL 15/03/2023 (Shift-01)

(a) 2.2 (b) 3.2
(c) 4.2 (d) 3.1

$(\sec x - \tan x) + (\operatorname{cosec} y + \cot y)$

$= \frac{1}{5} + \frac{3}{1}$

$= 0.2 + 3$

$= 3.2$

TYPE-V

- $\sin^2\theta + \cos^2\theta = 1$
- $2\sin^2\theta + 2\cos^2\theta = 2$
- $3\sin^2\theta + 3\cos^2\theta = 3$

65. If $10\sin^2\theta + 6\cos^2\theta = 7$, $0 < \theta < 90°$, then find the value of $\tan\theta$.

SSC CGL 24/09/2024 (Shift-03)

(a) $\sqrt{3}$ (b) 1

(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}$

$4\sin^2\theta + 6\sin^2\theta + 6\cos^2\theta = 7$

$\Rightarrow 4\sin^2\theta + 6 = 7$

$\Rightarrow 4\sin^2\theta = 1$

$\Rightarrow \sin^2\theta = \frac{1}{4}$

$\therefore \sin\theta = \frac{1}{2}$

$\theta = 30°$

$\tan 30° = \frac{1}{\sqrt{3}}$

$4S^2 = 3 + 3C$

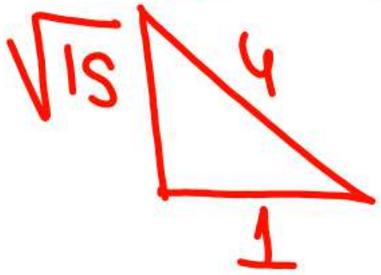
$\Rightarrow 4(1-C^2) = 3+3C$

$\Rightarrow 4 - 4C^2 = 3+3C$

$\Rightarrow 4C^2 + 3C - 1 = 0$

$-\frac{1}{4}$ $\quad +\frac{1}{4}$

$C = -1$ or $C = \frac{1}{4}$



66. If $4\sin^2\theta = 3(1+\cos\theta)$, $0° < \theta < 90°$, then what is the value of $\sqrt{15}\tan\theta + \frac{4}{\sqrt{15}}\sin\theta + 2\sec\theta$?

SSC CPO 28/06/2024 (Shift-02)

(a) 24

(b) $4\sqrt{15}$

(c) $\frac{8}{\sqrt{15}}$

(d) 8

$\sqrt{15} \times \frac{\sqrt{15}}{1} + \frac{4}{\sqrt{15}} \times \frac{\sqrt{15}}{4} + 2 \times \frac{4}{1}$

$= 15 + 1 + 8$

$= 24$



$S^2 = (1-C^2)$

$C^2 = (1-S^2)$

$x^2 + 5x + 6 = 0$

$-1/2 \quad -1/3$

$7c^2+5(1-c^2)=6$

$\Rightarrow 7c^2+5-5c^2=6$

$\Rightarrow 2c^2=1$

$\Rightarrow c^2=\frac{1}{2}$

$\therefore c=\frac{1}{\sqrt{2}}$ $\quad \therefore \theta=45^\circ$

67. If $7\cos^2\theta + 5\sin^2\theta - 6 = 0$, $(0^\circ < \theta < 90^\circ)$, then what is the value of $1+\sqrt{\frac{\sec\theta+\tan\theta}{\sec\theta-\tan\theta}}$?

SSC CGL 10/09/2024 (Shift-03)

(a) $\sqrt{2}+2$ (b) $\sqrt{2}+1$

(c) $\sqrt{2}-1$ (d) $\sqrt{2}-2$

$1+\sqrt{(\sec\theta+\tan\theta)(\sec\theta+\tan\theta)}$

$= 1+(\sec\theta+\tan\theta)$

$= 1+\sqrt{2}+1$

$= \sqrt{2}+2$

$S^2 - 3S + 2 = C^2$

$\Rightarrow S^2 - 3S + 2 = 1 - S^2$

$\Rightarrow 2S^2 - 3S + 1 = 0$

$+2 \quad -3 \quad +1$

$-2 \quad -1$

$S = 1 \qquad S = \frac{1}{2}$

$\theta = 90^\circ \qquad \theta = 30^\circ$

$\cos 60 - \sin 90 + \operatorname{cosec} 60^\circ$

$= \frac{1}{2} - \frac{1}{1} + \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{3 - 6 + 4\sqrt{3}}{6} = \frac{-3 + 4\sqrt{3}}{6}$

68. If $\frac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$, where $0^\circ < \theta < 90^\circ$, then what is the value of $(\cos 2\theta - \sin 3\theta + \operatorname{cosec} 2\theta)$

SSC CGL 21/04/2022 (Shift-02)

(a) $\frac{3 + 4\sqrt{3}}{6}$

(b) $\frac{-3 - 4\sqrt{3}}{6}$

(c) $\frac{3 - 4\sqrt{3}}{6}$

(d) ✓ $\frac{-3 + 4\sqrt{3}}{6}$

$\theta = 30^\circ$ (verified)

68. If $\dfrac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$, where $0^\circ < \theta < 90^\circ$, then what is the value of $(\cos 2\theta - \sin 3\theta + \operatorname{cosec} 2\theta)$

SSC CGL 21/04/2022 (Shift-02)

(a) $\dfrac{3+4\sqrt{3}}{6}$

(b) $\dfrac{-3-4\sqrt{3}}{6}$

(c) $\dfrac{3-4\sqrt{3}}{6}$

(d) $\dfrac{-3+4\sqrt{3}}{6}$

$+\frac{1}{2} - 1 + \frac{2}{\sqrt{3}}$

$+0.5 - 1$

$S \times C = 2C^2 - \frac{1}{4} C$

$\Rightarrow S = 2C^2 - \frac{1}{4}$

$\Rightarrow S = 2(1 - S^2) - \frac{1}{4}$

$\Rightarrow S = 2 - 2S^2 - \frac{1}{4}$

$\Rightarrow 2S^2 + S = \frac{7}{4}$

$\Rightarrow 8S^2 + 4S - 7 = 0$

$\Rightarrow \sin\theta = \frac{-4 + \sqrt{16 + 4 \times 8 \times 7}}{2 \times 8}$

$= \frac{\sqrt{240} - 4}{16}$

69. If θ is an acute angle and $\sin\theta \cos\theta = 2\cos^3\theta - \frac{1}{4}\cos\theta$, then the value of $\sin\theta$ is:

SSC CHSL 10/08/2023 (Shift-02)

(a) $\frac{\sqrt{15} - 1}{8}$

(b) $\frac{\sqrt{15} - 1}{4}$

(c) $\frac{\sqrt{15} + 1}{4}$

(d) $\frac{\sqrt{15} - 1}{2}$

$\frac{4\sqrt{15} - 4}{16} = \frac{\sqrt{15} - 1}{4}$

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$S = 2(1-S^2) - \frac{1}{4}$

$\Rightarrow 4S = 8 - 8S^2 - 1$

$\Rightarrow 8S^2 + 4S - 7 = 0$

$S = \frac{-4 + \sqrt{16 + 224}}{16}$

$= \frac{4\sqrt{15} - 4}{16}$

$= \frac{\sqrt{15} - 1}{4}$

69. If θ is an acute angle and $\sin\theta\cos\theta = 2\cos^2\theta - \frac{1}{4}$, then the value of $\cos\theta$, then the value of $\sin\theta$ is:

SSC CHSL 10/08/2023 (Shift-02)

(a) $\frac{\sqrt{15}-1}{8}$

(b) $\frac{\sqrt{15}-1}{4}$

(c) $\frac{\sqrt{15}+1}{4}$

(d) $\frac{\sqrt{15}-1}{2}$

70. **Find the value of θ, if $\sec^2\theta + (1-\sqrt{3})\tan\theta - (1+\sqrt{3}) = 0$, where θ is an acute angle.**

SSC CHSL 05/08/2021 (Shift-03)

(a) 30°
(b) 15°
(c) 45°
(d) 60°

(a) $\sec^2 30 + (1-\sqrt{3}) \times \frac{1}{\sqrt{3}} - (1+\sqrt{3})$

$= \frac{4}{3} + \frac{1}{\sqrt{3}} - 1 - 1 - \sqrt{3}$

(d) $4 + (1-\sqrt{3})\sqrt{3} - (1+\sqrt{3})$

$= 4 + \sqrt{3} - 3 - 1 - \sqrt{3}$

$= 0$

$$\frac{S^2}{C^2-3C+2}=1$$

$\Rightarrow S^2 = C^2-3C+2$

$\Rightarrow (1-C^2) = C^2-3C+2$

$\Rightarrow 2C^2-3C+1=0$

$1 \quad +2 \quad +1$

$2 \quad 2 \quad 2$

$C=0 \quad C=60$

71. If $\dfrac{\sin^2\theta}{\cos^2\theta-3\cos\theta+2}=1$, θ lies in the first quadrant, then the value of $\dfrac{\tan^2\frac{\theta}{2}+\sin^2\frac{\theta}{2}}{\tan\theta+\sin\theta}$ is:

SSC CGL TIER-II 03/02/2022

(a) $\dfrac{2\sqrt{3}}{27}$ (b) $\dfrac{5\sqrt{3}}{27}$

(c) $\dfrac{2\sqrt{3}}{9}$ (d) $\dfrac{7\sqrt{3}}{54}$

$$\frac{\tan^2 30+\sin^2 30}{\tan 60+\sin 60}=\frac{\frac{1}{3}+\frac{1}{4}}{\sqrt{3}+\frac{\sqrt{3}}{2}}=\frac{\frac{7}{12}}{\frac{3\sqrt{3}}{2}}=\frac{7}{18\sqrt{3}}\times\sqrt{3}=\frac{7\sqrt{3}}{54}$$

TYPE-VI

Bahubali Method

72. If $\sin A + \sin^2 A = 1$, then the value of $\cos^4 A + \cos^6 A$ is:

SSC CPO 04/10/2023 (Shift-01)

(a) 1
(b) cos A
(c) sin A
(d) 0

$S = 1 - S^2$

$\therefore S' = C^2$

$C^4 + C^6$

$= S^2 + S^3$

$= S(S + S^2)$

$= \sin A$

73. If $\cos A + \cos^2 A = 1$, then the value of $\sin^4 A + \sin^6 A$ is:

SSC CGL 26/07/2023 (Shift-01)

(a) 2
(b) cosA
(c) 1
(d) sinA

$C = 1 - C^2$

$\therefore C' = S^2$

$S^4 + S^6$
$= C^2 + C^3$
$= C(C + C^2)$
$= \cos A$

74. If $\cot^4\theta - \cot^2\theta = 1$, then the value of $\cos^4\theta + \cos^2\theta$ is:

SSC CGL 12/09/2024 (Shift-02)

(a) 2

(b) 1

(c) $\frac{1}{\sqrt{3}}$

(d) $\frac{3}{2}$

$\cot^4\theta = 1 + \cot^2 A$

$\Rightarrow \frac{C^4}{S^2} = \frac{1}{S^2}$

$\Rightarrow C^4 = S^2$

$\cos^4\theta + \cos^2\theta$

$= \sin^2\theta + \cos^2\theta$

$= 1$

75. If $\cot^4 A + \cot^2 A = 1$ then find the value of $\cos^4 A - 3\cos^2 A$.

***

SSC CHSL TIER-II 02/11/2023

(a) 2 (b) −1

(c) 1 (d) −2

$\cot^2 A(\cot^2 A + 1) = 1$

$\Rightarrow \cot^2 A \times \operatorname{cosec}^2 A = 1$

$\Rightarrow \frac{C^2}{S^2} \times \frac{1}{S^2} = 1$

$C^2 = S^4$

$(\cos^2 A)^2 - 3\cos^2 A$

$= (1 - S^2)^2 - 3C^2$

$= 1 + S^4 - 2S^2 - 3C^2$

$= 1 + C^2 - 2S^2 - 3C^2$

$= 1 - 2S^2 - 2C^2$

$= 1 - 2(S^2 + C^2)$

$= 1 - 2$

$= -1$

76. If $\sin^4 x + \sin^2 x = 1$, then what is the value of $\cot^4 x + \cot^2 x$?

SSC CHSL 17/03/2023 (Shift-02)

(a) – 2
(b) 2
(c) – 1
(d) 1

$S^4 = C^2$

$\Rightarrow S^2 = \frac{C^2}{S^2}$

$\sin^2 x = \cot^2 x$

$\sin^4 x + \sin^2 x = 1$

77. If $\tan^4\theta + \tan^2\theta = 1$, what is the value of $11(\cos^4\theta + \cos^2\theta)$?

SSC CGL 27/07/2023 (Shift-03)

(a) −11
(b) 8
(c) 0
(d) 11

$\tan^2\theta(\tan^2\theta+1)=1$

$\Rightarrow \tan^2\theta = \frac{1}{\sec^2\theta}$

$\Rightarrow \tan^2\theta = \cos^2\theta$

$11(\tan^4\theta+\tan^2\theta)$

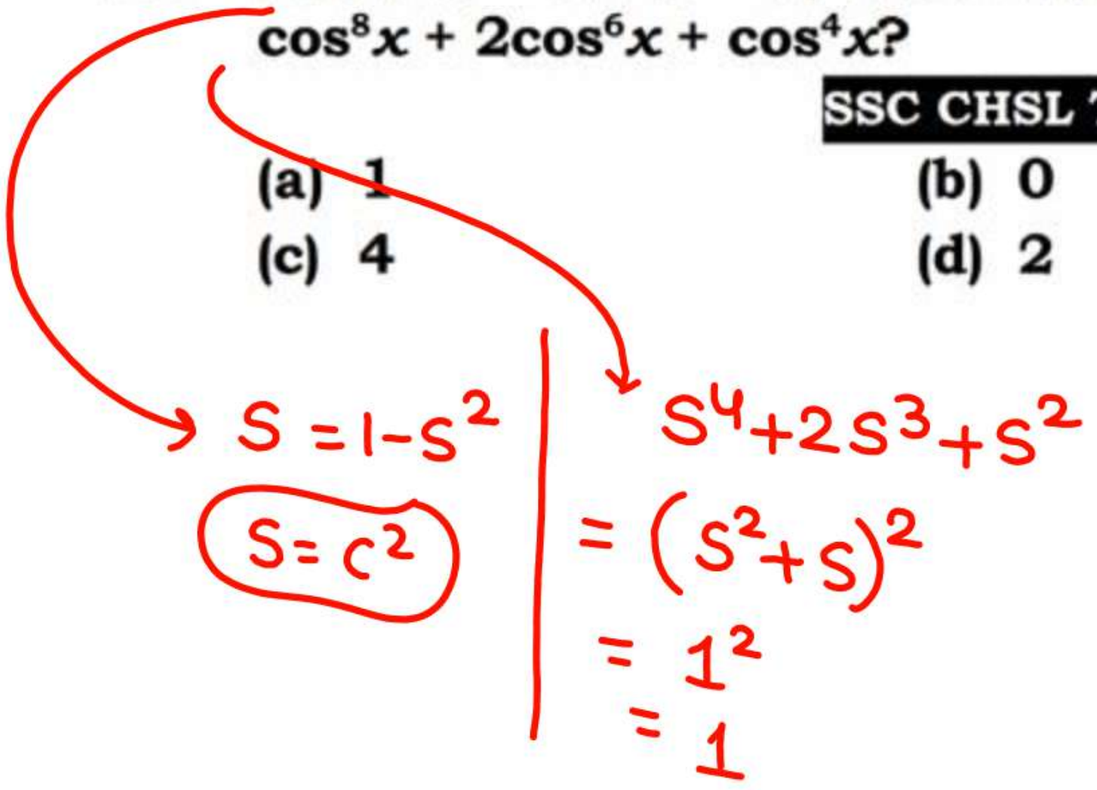
$= 11 \times 1$

$= 11$

78. If $\sin x + \sin^2 x = 1$, then what is the value of $\cos^8 x + 2\cos^6 x + \cos^4 x$?

SSC CHSL TIER-II 26/06/2023

(a) 1
(b) 0
(c) 4
(d) 2



79. If $\cos A = \sin^2 A$, and $a\sin^{12}A + b\sin^{10}A + c\sin^8 A + \sin^6 A = 1$, then $a + b + c = ?$

SSC CPO 11/11/2022 (Shift-01)

(a) 7
(b) 8
(c) 9
(d) 6

$c = s^2$

$a c^6 + b c^5 + c c^4 + c^3 = 1$

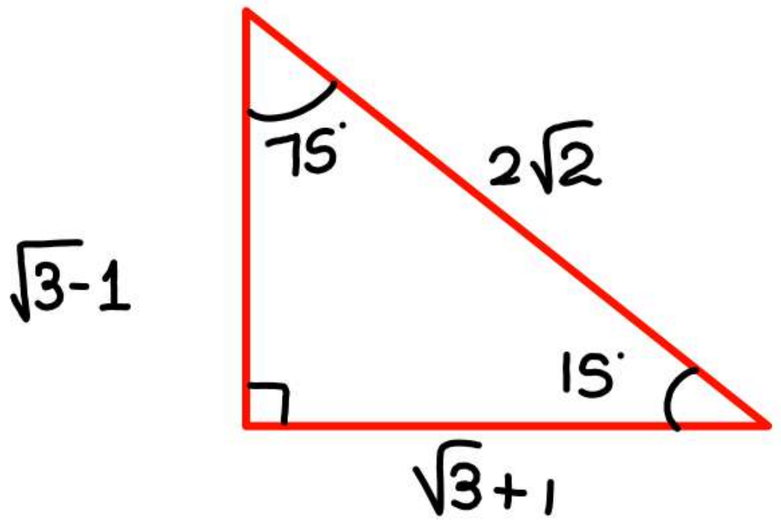
$= c^6 + 3c^5 + 3c^4 + c^3$

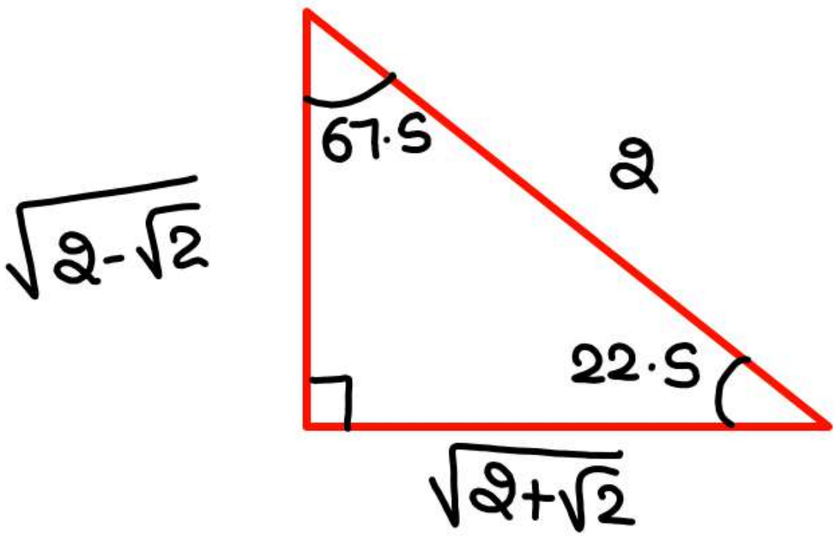
$= (c^2 + c)^3$

$= (c^2 + s^2)^3$

$= 1^3$

$= 1$


75°
2√2
15°
√3-1
√3+1


67.5°
2
22.5°
√2-√2
√2+√2

TYPE-VII

* $S - C = x$

$S + C = y$

Ans $(S-C)^2 + (S+C)^2 = x^2 + y^2$

$\Rightarrow S^2 + C^2 - 2SC + S^2 + C^2 + 2SC = x^2 + y^2$

$\Rightarrow 2 = x^2 + y^2$

$\therefore \sqrt{2 - x^2} = y$

* $S - C = 1$

$S + C = y$

Ans $y = \sqrt{2 - 1^2}$

$y = 1$

80. If $\sin\theta - \cos\theta = \frac{1}{5}$, then find the value of $\sin\theta + \cos\theta$

SSC CHSL 02/08/2023 (Shift-04)

(a) $\frac{5}{7}$

(b) $\frac{7}{5}$

(c) $\frac{5}{3}$

(d) $\frac{3}{5}$

$y = \sqrt{2 - \left(\frac{1}{5}\right)^2}$

$= \sqrt{2 - \frac{1}{25}} = \sqrt{\frac{49}{25}}$

$= \frac{7}{5}$

Q. $S - C = \frac{1}{3}$

$S + C = ?\ (y)$

Ans $y = \sqrt{2 - \left(\frac{1}{3}\right)^2}$

$= \sqrt{2 - \frac{1}{9}}$

$= \sqrt{\frac{17}{9}} = \frac{\sqrt{17}}{3}$

Q $S + C = \frac{1}{4}$

$S - C = ?\ (y)$

Ans $y = \sqrt{2 - \left(\frac{1}{4}\right)^2}$

$= \sqrt{2 - \frac{1}{16}} = \sqrt{\frac{31}{16}} = \frac{\sqrt{31}}{4}$

81. If $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$, then $(\cos\theta + \sin\theta)$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) $-\sqrt{2}\sin\theta$

(b) $\sqrt{2}\tan\theta$

(c) $\sqrt{2}\cos\theta$

(d) $-\sqrt{2}\cos\theta$

$C - S = \sqrt{2}\sin\theta$

$C + S = y$

$y = \sqrt{2 - 2\sin^2\theta}$

$= \sqrt{2(1 - \sin^2\theta)}$

$= \sqrt{2 \times \cos^2\theta}$

$= \sqrt{2}\cos\theta$

81. If $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$, then $(\cos\theta + \sin\theta)$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) $-\sqrt{2}\sin\theta$

(b) $\sqrt{2}\tan\theta$

(c) $\sqrt{2}\cos\theta$

(d) $-\sqrt{2}\cos\theta$

$C - S = \sqrt{2}\sin\theta$

$C + S = \sqrt{2}\cos\theta$

$C - S = \sqrt{2}\cos\theta$

$C + S = \sqrt{2}\sin\theta$

F-1

$$a\sin\theta + b\cos\theta = c$$

$$+ \quad b\sin\theta - a\cos\theta = d$$

$$a^2+b^2 = c^2+d^2$$

82. If $4\cos\theta + 3\sin\theta = x$ and $4\sin\theta - 3\cos\theta = y$, find the value of $x^2 + y^2$.

SSC CPO 11/11/2022 (Shift-01)

(a) 16 (b) 9

(c) 25 (d) 1

$3\sin\theta + 4\cos\theta = x$

$4\sin\theta - 3\cos\theta = y$

$3^2 + 4^2 = x^2 + y^2$

$25 = x^2 + y^2$

Q. $5\sin\theta + 12\cos\theta = 8$

$12\sin\theta - 5\cos\theta = ?$ (d)

Ans $5^2 + 12^2 = 8^2 + d^2$

$\Rightarrow 13^2 = 8^2 + d^2$

$\Rightarrow 105 = d^2$

$\Rightarrow d = \sqrt{105}$

Q. $7\sin\theta + 12\cos\theta = 9$

$12\sin\theta - 7\cos\theta = ?$

Ans $7^2 + 12^2 = 9^2 + x^2$

$\Rightarrow 112 = x^2$

$\Rightarrow 16 \times 7 = x^2$

$\Rightarrow 4\sqrt{7} = x$

83. If $\sin x + \cos x = \sqrt{2}\ \sin x$, then the value of $\sin x - \cos x$ is:

SSC CPO 28/06/2024 (Shift-02)

(a) $\sin x$

(b) $2\sin x$

(c) $-\sqrt{2}\ \cos x$

(d) $\sqrt{2}\ \cos x$

$S + C = \frac{\sqrt{11}}{3}$ (x)

$C - S = y$

$\therefore y = \sqrt{2 - \frac{11}{9}} = \sqrt{\frac{7}{9}}$

$= \frac{\sqrt{7}}{3}$

84. If $\sin\theta + \cos\theta = \frac{\sqrt{11}}{3}$, then the value of $(\cos\theta - \sin\theta)$ is:

SSC CPO 03/10/2023 (Shift-03)

(a) $\frac{\sqrt{5}}{3}$

(b) $\frac{7}{3}$

(c) $\frac{5}{3}$

(d) $\frac{\sqrt{7}}{3}$

Q. $a\operatorname{cosec}\theta + b\cot\theta = c$

$b\operatorname{cosec}\theta + a\cot\theta = d$

Ans $(a\operatorname{cosec}\theta + b\cot\theta)^2 - (b\operatorname{cosec}\theta + a\cot\theta)^2 = c^2 - d^2$

$\Rightarrow (a^2\operatorname{cosec}^2\theta + b^2\cot^2\theta + 2ab\operatorname{cosec}\theta\cot\theta) - (b^2\operatorname{cosec}^2\theta + a^2\cot^2\theta + 2ab\operatorname{cosec}\theta\cot\theta) = c^2 - d^2$

$\Rightarrow a^2(\operatorname{cosec}^2\theta - \cot^2\theta) + b^2(\cot^2\theta - \operatorname{cosec}^2\theta) = c^2 - d^2$

$\Rightarrow a^2 \times 1 + b^2(-1) = c^2 - d^2$

$\Rightarrow \boxed{a^2 - b^2 = c^2 - d^2}$

Q. $a\operatorname{cosec}\theta + b\cot\theta = c$

$b\operatorname{cosec}\theta + a\cot\theta = d$

$c^2 - d^2 = ?$

(a) $a^2 - b^2$

(b) $b^2 - a^2$

(c) $a^2 + b^2$

Q. $a\cot\theta + b\operatorname{cosec}\theta = c$

$b\cot\theta + a\operatorname{cosec}\theta = d$

$c^2 - d^2 = ?$

(a) $a^2 - b^2$

(b) $b^2 - a^2$

(c) $a^2 + b^2$

* $a\sin\theta + b\cos\theta = c$

$b\sin\theta - a\cos\theta = d$

$$a^2 + b^2 = c^2 + d^2$$

* $a\operatorname{cosec}\theta + b\cot\theta = c$

$b\operatorname{cosec}\theta + a\cot\theta = d$

Soln

85. If $a\cot\theta + b\operatorname{cosec}\theta = p$ and $b\cot\theta + a\operatorname{cosec}\theta = q$ then $p^2 - q^2$ is equal to _____.

SSC CGL TIER-II 07/03/2023

(a) $b^2 - a^2$
(b) $a^2 - b^2$
(c) $b - a$
(d) $a^2 + b^2$

Ans : $b^2 - a^2$

Q. $7\operatorname{cosec}\theta + 5\cot\theta = 8$

$5\operatorname{cosec}\theta + 7\cot\theta = y$

Ans

$$7^2 - 5^2 = 8^2 - y^2$$

$$\Rightarrow \quad 24 = 64 - y^2$$

$$\Rightarrow \quad y^2 = 40$$

$$\Rightarrow \quad y = 2\sqrt{10}$$

Q. $a\sin\theta + b\cos\theta = x$

$b\sin\theta - a\cos\theta = y$

$a^2+b^2 = x^2+y^2$

Q. $a\sec\theta \pm b\tan\theta = x$

$b\sec\theta \pm a\tan\theta = y$

$a^2 - b^2 = x^2 - y^2$

Q. $\sin\theta + \cos\theta = \sqrt{2}\cos\theta$

$\sin\theta - \cos\theta = (y)$

Ans

$1^2 + 1^2 = 2\cos^2\theta + y^2$

$\Rightarrow 2 - 2\cos^2\theta = y^2$

$\Rightarrow 2(1-\cos^2\theta) = y^2$

$\Rightarrow \sqrt{2}\sin\theta = y$

